
Physics-R1: An Audited Olympiad Corpus and Released Verifiers for Visual Physics Reasoning

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Abstract

Trackable improvement in multimodal physics reasoning rests on a training-and-evaluation system that is itself rarely verified: the corpora a model trains on, the reward it is optimized against, and the benchmarks and judges that score it. We audit this system end to end and find that standard construction practices systematically distort measurement: contamination slips past n -gram deduplication, translation degrades problems, saturated multiple-choice formats overstate capability, partial-credit training rewards are easier to exploit than to earn, and open-ended grading silently depends on the choice of judge. Left unverified, these distortions inflate reported progress and leak test knowledge into training. We answer with a released verifier system: a three-stage contamination audit that certifies the train/test boundary behind an audited multimodal training corpus and a held-out olympiad benchmark; a binary answer verifier that supplies the reinforcement-learning training reward; and an answer-judging harness that brackets every open-ended score between a deterministic strict layer and a large-language-model liberal layer. All per-record verdicts are released and cross-checked against an independent open-weight judge, whose substitution shifts absolute scores but preserves the sign of every base-to-trained lift. Training against the binary answer verifier confirms the certified corpus supports training: a reference recipe lifts an 8B open-source base by 18.3 points on the held-out benchmark across three seeds. The simple binary reward also beats a dense partial-credit variant on three of four open-ended benchmarks, tying the fourth. All verifiers, verdicts, and datasets are public.

1 Introduction

Physics olympiad problems are among the most demanding tests of multimodal scientific reasoning: solving one means reading a diagram, selecting the governing physical law, and carrying symbolic and numerical work through to a checkable final answer. As vision-language models (VLMs) post rapidly rising scores on physics benchmarks, those scores increasingly stand in for claims about scientific reasoning ability and steer what gets trained next. The challenge is that reliably improving a VLM’s physics reasoning rests on an entire verifiable training-and-evaluation system: the training data, the training reward verifier, and a valid benchmark with a fixed judge. Each component is a place where measurement can silently break, yet only the final score is usually inspected. Five common practices undermine the system from within. Training pools aggregate public problem collections and are deduplicated against evaluations by a single n -gram pass; paraphrases survive it, so a model may be scored on lightly reworded versions of its own training data, and recall reads as reasoning. Benchmarks are evaluated in English even when their problems were composed in another language, as if translation were free; wording, idiom, and notation shift in translation, and the score shifts with them. Multiple-choice question (MCQ) formats compress scores against the frontier ceiling, where guessing floors and option elimination do part of the work, so differences in genuine capability stop registering. In reinforcement-learning post-training, rewards shaped with

partial credit pay out for answer form as well as physics, and a policy under optimization pressure collects the form. And when answers are open-ended, the grade depends on the judge itself: which model grades, under what prompt, at what leniency. That instrument is almost never held fixed or released. We therefore treat the system itself as the object to be verified: this paper audits every component, states each failure as a numbered finding, and releases the verifiers and artifacts constructed to close each gap (Figure 1).

Auditing every component yields five findings, from the training data to the evaluation judge. **Finding 1 (contamination):** a standard single-stage 5-gram Jaccard audit passes public physics training pools as clean against six public evaluations. Our three-stage audit of n-gram overlap, then embedding cosine, then a large language model (LLM) judge surfaces 134 near-duplicates of evaluation problems in SciInstruct alone (§3.1). **Finding 2 (translation):** on 59 paired Estonian/English versions of identical olympiad problems, the same frontier model scores 17 percentage points (pp) higher on the Estonian originals (§5.2). **Finding 3 (format saturation):** identical model weights score 46 points lower on our novel-source open-ended olympiad evaluation than on a public MCQ benchmark, a spread produced by evaluation format, problem genre, and source novelty on fixed weights (§5.2). **Finding 4 (reward robustness):** training against the binary answer verifier beats a dense partial-credit variant on three of four open-ended benchmarks and ties the fourth, by up to 8.9 pp at matched training steps; under optimization pressure, shaped partial credit rewards format proxies rather than physics (§5.3). **Finding 5 (judge reliability):** re-scoring every re-derivable open-ended column with a local open-weight judge shifts absolute scores by up to +37 pp and can reverse cross-model rankings, so comparisons are meaningful only under a fixed, released judge (§6).

The release is organized as a single verifiable system (Figure 1): a certified data boundary, training that optimizes against a verifier, and measurement through a released judging harness. Three deployed verifiers anchor it. Verifier 1 is the three-stage contamination audit: it certifies a train/test boundary between any training pool and any evaluation, and ships as runnable code with per-pair verdicts (§3). Its certified artifacts are PHYSORP-A, a 6,432-record audited multimodal corpus; PHYSR1CORP, a 2,268-record closed-form pool for reinforcement learning (RL); and PHYSOLYM-A, a 500-problem held-out open-ended olympiad benchmark with organizer difficulty labels and a bilingual English/Estonian subset (§3.3). Verifier 2 is the binary answer verifier whose verdict is the RL training reward (§4). Verifier 3 is the answer-judging harness that scores open-ended physics answers: a deterministic strict layer, plus an LLM liberal layer whose per-record verdicts are released (§5.1).

The final artifact, Physics-R1, is a reference RL recipe, included as evidence that the certified corpus supports training rather than as a method contribution; it closes the loop on all three verifiers, training inside the boundary Verifier 1 certifies, optimizing against Verifier 2, and scored by Verifier 3. Cold-started from an open-source 8B VLM with a binary correctness reward, it lifts the base by 18.3 pp on PHYSOLYM-A at the 3-seed mean (§5). Only the strongest frontier model remains ahead of it, by 7.1 pp.

2 Related Work

Rule-based RL for reasoning. DeepSeek-R1 [DeepSeek-AI, 2025] established that simple rule-based rewards suffice to train competitive math reasoners, via GRPO [Shao et al., 2024]; MM-Eureka [Meng et al., 2025] extended the recipe to VLMs with a difficulty curriculum, DAPO [Yu et al., 2025] added decoupled clipping and dynamic sampling, and GSPO [Zheng et al., 2025] moved to sequence-level importance weighting. Physics-R1 inherits these components unchanged. Although physics steps carry units and conservation laws that admit per-step verification, a binary reward is the better default under group-normalized advantages (§4); the dense physics-native reward is an ablation.

Physics QA benchmarks. PhyX [Shen et al., 2025], OlympiadBench-Physics [He et al., 2024], UGPhysics [Xu et al., 2025], PhysReason [Zhang et al., 2025], MMMU/MMMU-Pro [Yue et al., 2024, 2025], MMK12 [Meng et al., 2025], PHYBench [Qiu et al., 2025], and PhysUniBench [Wang et al., 2025b] are the canonical references. Only PHYBench, OIBench, and PutnamBench publish a contamination protocol. None publish the three-stage pairwise audit of n-gram overlap, embedding

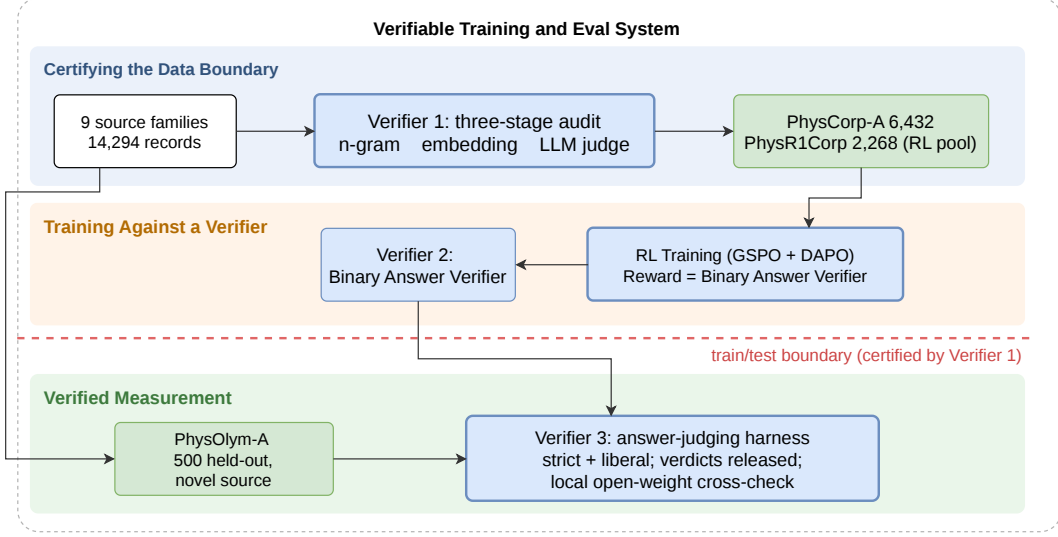


Figure 1: **The verifiable training-and-evaluation system overview.** Verifier 1, the three-stage audit, certifies the corpora and the held-out benchmark on opposite sides of the train/test boundary. Training optimizes against Verifier 2, the binary answer verifier, a deterministic oracle of the same family as Verifier 3’s strict layer. All reported numbers flow through Verifier 3, the answer-judging harness, whose per-record verdicts are released and cross-checked by a local open-weight judge (Appendix D).

cosine, and LLM judgment that we introduce in §3.1. Table 9 (Appendix H.6) maps our released artifacts against related benchmarks on nine axes.

Contamination audits and other prior work. PutnamBench [Tsoukalas et al., 2024], FrontierMath [Glazer et al., 2024], HLE [Phan et al., 2025], and EnigmaEval [Wang et al., 2025a] provide release-policy templates and dismissal grounds; methodological work spans n-gram audits [Sainz et al., 2023], the rephrased-samples failure mode [Yang et al., 2023] (which our Stage 2 catches), contamination-impact measurement [Singh et al., 2024], and performance-based detection [Dekoninck et al., 2024]; the survey of Ravaut et al. [2025] consolidates these. We import the math template, adding the embedding-cosine pass because physics statements (units, vectors, figure references) are more paraphrase-sensitive than typical math problems—a sensitivity Table 3 quantifies. PhysBench [Chow et al., 2025] evaluates physical-world understanding over interleaved video-image-text tasks, an orthogonal scope. Multilingual benchmarks have proliferated [Xuan et al., 2025, Ahuja et al., 2024, Wu et al., 2025]; our cross-lingual finding (§5.2) instead evaluates 59 identical problems in original Estonian and in English translation on the same model with paired tests, isolating a within-problem effect.

3 The Verifiable System I: Certifying the Data Boundary

The system’s first stage certifies its data boundary (Figure 1). Verifier 1, the released three-stage contamination audit, performs the certification: its input is any pair of problem pools; its output is a verdict for every candidate overlap, plus the certified-disjoint pools. This section describes the verifier’s construction and measured failure modes (§3.1), then the artifacts it certifies: PHYSCORP-A (§3.2), PHYSR1CORP, and the held-out PHYSOLYM-A (§3.3). All artifacts carry per-source licenses and Croissant metadata (Appendix E); release and replay details are in Appendix F.

3.1 Construction and Failure Modes

The verifier applies one definition of contamination pairwise between a training pool, four external corpora, and the held-out splits. *Stage 1 (n-gram recall)*: tokenize each problem statement, build

Table 1: **A standard single-stage audit calls every public training pool clean; the three-stage audit surfaces 134 near-duplicates of evaluation problems and certifies both released pools clean.** Rows are public physics eval splits; columns are training pools. Each cell reports Stage-1 / Stage-2 / Stage-3 pair counts under Algorithm 1: Stage-1 is 5-gram Jaccard ≥ 0.4 , Stage-2 is embedding cosine ≥ 0.85 , and Stage-3 is the LLM-judge precision filter separating close duplicates from same-topic neighbors (analysis in §3.1).

Eval ↓ / Train pool →	Other published pools (we re-audit)			This work (cleaned)	
	UGPhysics-Train (200 sub)	SciInstruct (en_phy_chem; 42 K)	MMK12 (MM-Eureka; 15 K)	PHYSCORP-A (6,432)	PHYSR1CORP (2,268)
PHYSOLYM-A (500)	0 / 0 / 0	0 / 163 / 8	0 / 0 / 0	0 / 0 / 0	0 / 3 / 0
PhyX-1k (1,000)	0 / 1 / 0	0 / 703 / 46	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
MMMU-Pro Phys (60)	0 / 0 / 0	0 / 141 / 7	0 / 0 / 0	0 / 0 / 0	0 / 1 / 0
OlymBench (692)	0 / 2 / 0	0 / 130 / 15	0 / 0 / 0	0 / 0 / 0	0 / 4 / 0
PhysReason-full (1,200)	0 / 4 / 0	0 / 2,687 / 36	0 / 62 / 0	0 / 0 / 0	0 / 11 / 0
PhysUniBench-en (1,022)	0 / 2 / 0	0 / 1,027 / 22	0 / 4 / 0	0 / 0 / 0	0 / 0 / 0
<i>Total S2 / S3 real</i>	9 / 0	4,846 / 134	66 / 0	0 / 0	19 / 0

Format: **Stage-1 / Stage-2 / Stage-3** pair counts (Stage-3 = Haiku-4.5 LLM-judge classifying each Stage-2 candidate as close duplicate vs. same-topic neighbor). Per-eval cells attribute a pair to every eval split containing the matched item, so the SciInstruct cells sum to 4,851; the Total row counts the 4,846 unique judged pairs.

its 5-gram shingle set, and flag pairs with Jaccard overlap $J \geq 0.4$. *Stage 2 (paraphrase recall)*: embed each statement with `mxbai-embed-large` and flag pairs with cosine similarity ≥ 0.85 ; this catches the reworded duplicates Stage 1 misses, but also flags problems that merely share a topic. *Stage 3 (precision filter)*: an LLM judge (Haiku 4.5) reads both statements of each Stage-2 candidate and classifies the pair as a *close duplicate* (a paraphrase or numeric variation of the same problem) or a *same-topic neighbor* (related physics, distinct setup). Only close duplicates are removed. Pseudocode is in Algorithm 1 and worked examples in Appendix A.

Table 1 shows why all three stages are needed (Finding 1). The three public training pools we re-audit all pass Stage 1 with zero hits against all six public evals, yet Stage 2 surfaces 4,846 close-content candidate pairs in SciInstruct alone. Of these, Stage 3 confirms 134 as near-duplicates of evaluation problems. The close-duplicate rate is sharply cosine-driven (100% at $\cos \geq 0.95$ down to 1.5% at the threshold edge; Appendix A, Table A): cosine provides the recall, the LLM judge the precision. PHYSCORP-A retains zero Stage-2 candidates by construction; PHYSR1CORP drops 165 near-duplicates (Appendix A.1), and its 19 remaining Stage-2 candidates are all same-topic neighbors.

Measured failure modes of Verifier 1. A verifier is only trustworthy if its own failure modes are measured, so we re-audited a 1,679-record sample of the released pre-audit pool, a sample curated under the field-default workflow of 5-gram deduplication. Against an internal 500-record evaluation, the single-stage audit detects a 3.3% leak rate, all exact copies. Adding the embedding stage raises the detected rate to 8.8%; the gap is paraphrased leakage that no n-gram check can see. The detected rate is also threshold-sensitive, sweeping 4.7–27.1% as the cosine cutoff moves from 0.90 to 0.80 (Appendix A.1, Table 3), which is why released verdicts always ship with their operating thresholds. Stage-3 precision rests on the LLM judge plus manual spot checks; a human-labeled validation on a cosine-stratified sample is left to follow-up work.

3.2 The Certified Corpora

The corpus is drawn from nine source families (Table 10, Appendix H.7). Five families are repackaged from existing collections under documented licenses: UGPhysics [Xu et al., 2025], OpenStax College and University Physics [OpenStax, 2024], Physics Stack Exchange [Stack Exchange Inc., 2024], an MMMU seed with o1 chain-of-thought (CoT) traces [Yue et al., 2024], and PhysReason [Zhang et al., 2025]. Four families contribute material new to machine-readable form. The Estonian Physics Olympiad collection [Estonian Physics Olympiad, 2018] contributes 418 problems spanning 2004–2018, each carrying an organizer-issued 1–10 difficulty label. Of these, 201 carry official English translations issued by the olympiad organizers, forming the bilingual English/Estonian (EN/ET) subset; the remaining 217 exist only in Estonian. We deliberately do not machine-translate the Estonian-only problems: that would inject into our own release exactly the confound Finding 2 documents. Kevin Zhou’s olympiad handouts [Zhou, 2024] contribute 692 problems with native

1–5 point values, with inline secondary-source attribution preserved per record (Appendix E). Refreshed scrapes of six international olympiads (IPhO, NBPhO, EuPhO, APhO, USAPhO, INPhO) complete the pool [IPhO Unofficial Archive, 2025, NBPhO Committee, 2025, EuPhO Committee, 2025, APhO 2025 Organizing Committee, 2025, American Association of Physics Teachers, 2025, Homi Bhabha Centre for Science Education, 2025]. Per-source licenses are listed in Appendix E and carried through to each record.

PHYSCORP-PRE-AUDIT ships in full, so downstream users can re-run the verifier from scratch. PHYSCORP-A is the 6,432-record subset that survives all three audit stages plus a re-audit against PhysReason-full and PhysUniBench-en; that re-audit drops a further 804 records, mostly against PhysReason-full. The released pool is disjoint, at the joint operating thresholds, from all six public evaluations we audit against, including the held-out PHYSOLYM-A. The candidate-to-release cleanup for PHYSR1CORP is described in §3.1.

Approximately 73 PHYSR1CORP records (3.2%) have LLM-touched problem statements, documented per-distribution in the Croissant metadata; PHYSOLYM-A contains no synthetic problem content.

3.3 PHYSOLYM-A: The Certified Held-Out Benchmark

Standard physics-VL benchmarks no longer resolve frontier-class differences. PhyX clusters top entries within ten points of its 80% ceiling; OlympiadBench-Physics predates the contamination-audit discipline; UGPhysics is itself a candidate for audited training data rather than held-out evaluation. Physics-R1’s headline evaluation and reward ablation depend on a held-out signal that is non-saturating and contamination-clean against the training pool.

PHYSOLYM-A (Physics Olympiad, Audited) contains 500 problems: 200 from Kevin Zhou’s handouts, 136 from the Estonian Physics Olympiad collection, 85 from an IPhO/NBPhO/EuPhO scrape, and 79 from an APhO/USAPhO/INPhO scrape. 499 of the 500 are novel-source under our four-corpus audit. Native difficulty signals are preserved: 131 of the 136 Estonian records carry organizer-issued 1–10 difficulty labels, and 38% of all records carry Zhou’s 1–5 point values. Verifier 1 certifies zero Stage-3 near-duplicate overlaps between the audited training pool and PHYSOLYM-A. The single non-novel record, an EuPhO 2020 problem shared with OlympiadBench-Physics, is disclosed in Appendix A rather than silently dropped. Scoring is performed by Verifier 3 (§5.1).

4 The Verifiable System II: Training Against a Verifier

The system’s second stage trains inside the boundary that Stage I certifies, optimizing against Verifier 2, the binary answer verifier. Physics-R1, the reference recipe instantiating this stage, is reported as evidence that the certified corpus has training utility; it is not an algorithmic contribution. The optimizer is GSPO [Zheng et al., 2025] with DAPO’s decoupled clipping and dynamic sampling [Yu et al., 2025], both unmodified. For each prompt x the policy samples $K=16$ rollouts $\{y_k\} \sim \pi_{\theta_{\text{old}}}(\cdot | x)$, each rollout is scored by a scalar reward $r(y_k, x)$, and advantages are normalized within the group:

$$A_k = \frac{r(y_k, x) - \bar{r}}{\sigma_r + \varepsilon}, \quad w_k(\theta) = \left(\frac{\pi_{\theta}(y_k | x)}{\pi_{\theta_{\text{old}}}(y_k | x)} \right)^{1/|y_k|}, \quad (1)$$

$$\mathcal{L}_{\text{GSPO}} = -\mathbb{E} \left[\frac{1}{K} \sum_k \min(w_k A_k, \text{clip}(w_k, 1 \pm \epsilon) A_k) \right] + \beta_{\text{KL}} D_{\text{KL}}(\pi_{\theta} \| \pi_{\text{base}}),$$

with $(\bar{r}, \sigma_r)$ the group mean and standard deviation and π_{base} the Qwen3-VL-8B-Thinking base checkpoint [Qwen Team, 2025]. Training cold-starts from the base model and keeps a small KL anchor to it. Prompts the policy already always or never solves are filtered out by the MM-Eureka difficulty curriculum [Meng et al., 2025], which removes roughly 22% of prompts. Early stopping uses a held-out multiple-choice signal rather than open-ended judging. The full configuration—context budgets, clipping thresholds, hardware, and the training loop in pseudocode—is consolidated in Appendix F (Algorithm 3, Table 11).

Two reward shapes: binary (recommended) vs. dense (ablation). Physics rollouts expose per-step signals (units, conservation laws, symbolic form), so a denser reward looks free. We compare:

$$\begin{aligned} \text{(binary, recommended)} \quad r_{\text{bin}}(y, x) &= \mathbb{I}[\text{MATCH}(\text{EXTRACTBOXED}(y), g(x))] \in \{0, 1\}, \\ \text{(dense, ablation)} \quad r_{\text{dense}} &= \text{clip}(r_{\text{ans}} + r_{\text{fmt}} + r_{\text{dim}} + r_{\text{sym}} + r_{\text{cons}}, -1, 1). \end{aligned} \quad (2)$$

Here MATCH is Verifier 2 of Figure 1, the binary answer verifier: a deterministic oracle of the same family as Verifier 3’s strict layer (§5.1). It accepts a multiple-choice letter match, a numeric match within $\pm 1\%$ tolerance, or symbolic equivalence (Appendix C.1). The recipe therefore optimizes directly against verifier verdicts. The dense variant adds four bounded shaping terms for answer formatting, dimensional consistency, symbolic well-formedness, and conservation-law violations (Appendix C.2). Empirically, binary wins on three of the four non-MCQ benchmarks and ties the fourth (Finding 4; per-benchmark margins in §5.3). The theoretical account (properties P1–P3, Appendix C.1) shows that shaping matters only when it reorders rollouts within a group, and those reorderings concentrate among wrong answers, rewarding format proxies rather than physics. We ship binary as the deployable default: under optimization pressure a partial-credit verifier invites reward hacking, while the binary verifier is robust to being trained against.

5 The Verifiable System III: Verified Measurement

The system’s final stage is measurement, and every reported number flows through a released verifier. §5.1 describes Verifier 3, which scores every open-ended column of Table 2. §5.2 characterizes PHYSOLYM-A as a measurement instrument, grounding Findings 2–3 (Finding 1 is grounded in §3.1); §5.3 reports Physics-R1 results and the reward ablation grounding Finding 4; Finding 5 is grounded in §6 (the seed-42 row of Table 2 is the released reference checkpoint; the 3-seed row, the headline result, aggregates three seeds).

5.1 Verifier 3: The Answer-Judging Harness

The harness verifies candidate answers against olympiad gold at two levels of stringency. Its strict layer is deterministic (multiple-choice letter, numeric-tolerance, or symbolic match) and involves no LLM. Its liberal layer uses an LLM judge (Sonnet 4.5 [Anthropic, 2025]) with one YES/NO call per gold answer: for multi-part problems on PhysReason and PhysUniBench open-ended (PUB-OE), the judge is called once per gold sub-answer, and a problem counts only if every sub-part is judged correct. OlympiadBench-Physics and PHYSOLYM-A receive a single call per problem against the full gold solution. All open-ended columns of Table 2 report *problem-level* liberal accuracy under this harness.

The harness is fully auditable: judge scripts with verbatim prompts, per-record verdicts (`judge_audit.json`), and run policies are all released (Appendix D), so every reported cell can be re-derived without re-querying any API. The 13.9% unjudgeable rate on PHYSOLYM-A stems from gold solutions that are grading rubrics or figure-only references. Judge optimism is bounded three ways: a 4.7 pp strict-vs.-liberal gap on PHYSOLYM-A, two-seed Cohen’s κ [Cohen, 1960], and a 100-problem human-graded subset (Appendix D); judge-choice calibration is in §6.

5.2 PHYSOLYM-A as a Measurement Instrument

Same-model evaluation reveals a 46-point format-and-novelty gradient (Finding 3). On identical Sonnet 4.5 weights evaluated in the same week, the score sweeps from 79.7% on PhyX (4-way MCQ) down to 50.4% liberal on OlympiadBench-Physics and 33.4% liberal on PHYSOLYM-A. Format, genre, and novelty together move the score by 46 points on fixed weights, the strongest evidence for Finding 3. Three forces drive the drop: format (MCQ vs. open-ended), genre (K-12/early-undergraduate vs. competition-grade), and contamination removal (only PHYSOLYM-A is three-stage audited). The PhyX→OlympiadBench step, roughly 29 pp, is dominated by format and genre; the OlympiadBench→PHYSOLYM-A step, roughly 17 pp, by audit and novelty. On the related OlympiadBench-Physics split, electromagnetism is hardest and astrophysics easiest (Appendix H.1).

Difficulty-stratified accuracy from organizer-issued labels. The Estonian problems carry organizer-issued 1–10 difficulty labels, eliminating self-annotation circularity. On the 131 judge-

able problems among the 136 with labels, Sonnet 4.5 strict accuracy decays near-monotonically from 58.8% at difficulty 1 to a 0% floor in four bins (Appendix H.7, Table 13); even the easiest bin scores below Sonnet’s PhyX average, and the clean zeros mark the benchmark as non-saturating.

Cross-lingual ablation: 17-point translation delta on identical problems (Finding 2). The bilingual subset enables a controlled experiment on 59 paired problems graded against the same gold by the same judge. Sonnet 4.5 scores 30.5% strict on the Estonian originals but 13.6% on the English translations (sign test $p=0.011$; further tests in Appendix H.4). Disagreements are asymmetric: 13 correct in Estonian but wrong in English, 3 the reverse. For models with weak Estonian coverage the gap is expected to flip in sign, which we pre-register as a follow-up (Appendix H.4, item viii).

5.3 Physics-R1

Recipe. Physics-R1 is the recipe of §4 cold-started from Qwen3-VL-8B-Thinking [Qwen Team, 2025] on PHYSR1CORP with a binary correctness reward; a 1,000-problem audit-clean MCQ subset of PhyX [Shen et al., 2025] (PhyX-1k in Tables 2 and 1), certified disjoint from the pool, is the in-training early-stop signal.

Table 2: **The binary seed-42 checkpoint lifts the 8B base in every column, and the largest lift lands on the audited held-out olympiad benchmark.** All open-ended columns use problem-level liberal accuracy under Verifier 3’s released Sonnet judges (per-sub-answer protocol for PhysReason and PUB-OE, problem-level for the olympiad columns; Appendix D); every sub-question of a multi-part problem must be judged correct for the problem to count. LLaVA-OneVision’s open-ended cells are local-judge scores (see the § footnote). The MCQ random baseline is 25%. **Bold / underline:** best / second-best per column across the open-source and this-work blocks. The Sonnet PhysReason cell ([†]) is regenerated at the 16,384-token budget used by Physics-R1 and all open-source baselines. GPT-4o PhyX cells are from Shen et al. [2025]; Gemini PhyX cells (*) are measured here. Local-judge re-scores of every re-derivable column: §6 and Appendix D.

Model	MCQ		Open-ended (problem-level, liberal Sonnet judge)			
	PhyX-1k	PhyX-3k	PhysReason	PUB-OE	OlymBench	PHYSOLYM-A
<i>Closed-source frontier</i>						
Claude Sonnet 4.5	79.7	80.6	49.1 [†]	25.4	50.4	33.4
Gemini 2.5 Pro	75.1*	49.8*	38.8	33.4	37.4	12.2
GPT-4o	70.4	53.6	51.1	31.0	19.7	19.5
<i>Open-source bases</i>						
Qwen3-VL-32B-Thinking	73.8	84.2	25.1	32.8	53.9	13.2
Qwen3-VL-8B-Thinking (base)	73.7	74.4	23.9	35.3	39.3	8.0
InternVL3-8B	46.8	43.1	13.3	23.5	10.4	4.0
LLaVA-OneVision-7B [§]	37.4	36.0	3.5	19.2	8.4	7.2
<i>This work (subscripts: Δ vs. Qwen3-VL-8B-Thinking base)</i>						
Physics-R1 (dense)	78.3 _{+4.6}	<u>77.5</u> _{+3.1}	23.3 _{-0.6}	37.7 _{+2.4}	40.5 _{+1.2}	19.2 _{+11.2}
Physics-R1 (binary, seed 42)	<u>78.0</u> _{+4.3}	76.9 _{+2.5}	<u>32.2</u> _{+8.3}	<u>37.0</u> _{+1.7}	45.4 _{+6.1}	<u>25.6</u> _{+17.6}
Physics-R1 (binary, 3-seed mean $\pm \sigma$) [‡]	77.8 _{+4.1} <u>± 0.3</u>	76.9 _{+2.5} <u>± 0.3</u>	39.6 _{+15.7} <u>± 6.4</u>	34.5 _{-0.8} <u>± 3.6</u>	46.2 _{+6.9} <u>± 1.5</u>	26.3 _{+18.3} <u>± 1.7</u>

[‡] 3-seed mean over seeds {42, 17, 23} trained on the audited PHYSR1CORP (2,268 records, binary reward); per-seed values and the lift decomposition are in Appendix H.7. [§] LLaVA-OneVision: PhyX cells are standard multimodal MCQ exact-match; open-ended cells report local-judge liberal accuracy (the local judge is more lenient than Sonnet, so Sonnet-judged values would be lower); strict scores and protocol caveats in Appendix D.

Capability across formats. Table 2 reports Physics-R1 alongside closed-frontier and open-source baselines on three answer formats and the held-out olympiad split, including InternVL3-8B [Zhu et al., 2025a] and the LLaVA-OneVision-7B distinct-lineage baseline [Li et al., 2024]; a local open-weight judge re-score of every re-derivable open-ended column is reported in §6. The 8B base attains 73.7% on PhyX-1k and its 32B sibling is indistinguishable there (73.8%), so scale alone does not move a saturated MCQ benchmark; all evaluations allow the full 16,384-token CoT budget (Appendix H.7).

The binary checkpoint lifts the base most where the eval is audited and novel. The largest lift lands on PHYSOLYM-A liberal: +18.3 pp at the 3-seed mean (per-column lifts are the subscripts in Table 2). Lifts on the saturated public open-ended splits are much smaller (the 3-seed PUB-OE mean dips 0.8 pp below base), exactly as Findings 1 and 3 predict: where the base is already near its ceiling there is little headroom, and where the eval is novel-source and audited the post-training lift is large. Across seeds the lift is reproducible (per-seed values in Appendix H.7); PHYSOLYM-A liberal averages 26.3 ± 1.7 . PhysReason averages 39.6 ± 6.4 , with seed 42 an outlier roughly 11 pp below the other seeds, so we report all three seeds rather than the mean alone.

Where Physics-R1 helps: base failure modes it mitigates. The PHYSOLYM-A lift corresponds to roughly 92 problems flipping from wrong to correct. Hand-inspecting 30 flips reveals three recurring base failure modes, each addressed by a recipe lever: not committing to a boxed final answer (fixed by the binary reward); dimensionally-consistent but answer-wrong shortcuts (mitigated by the difficulty curriculum); and failure to integrate evidence across multiple images (mitigated by preserving the base visual encoder). Physics-R1 does *not* fix genuine physics-content gaps: graduate-level perturbation-theory problems remain wrong on both checkpoints (transcripts in Appendix H.1).

PHYSOLYM-A grounds the training-utility claim. On PHYSOLYM-A liberal, the 3-seed mean exceeds every open-source baseline and two of the three closed frontier models we test (Table 2); only Sonnet 4.5 remains ahead, by 7.1 pp.

Reward-shape ablation: binary wins on open-ended formats (Finding 4). At matched step 60 on seed 42, the dense reward slightly leads on saturated MCQ, by at most 0.6 pp. Binary leads dense by 8.9 pp on PhysReason. It leads by 4.9 pp on OlympiadBench-Physics and by 6.4 pp on PHYSOLYM-A liberal. On PUB-OE the two effectively tie, with dense ahead by 0.7 pp. The binary advantage concentrates on the multi-part numerical column and the audited held-out column. Per-component reward drop-out (Appendix H.7, Table 12) is left to follow-up work.

6 Discussion, Limitations, and Conclusion

Robustness of the verifiers (Finding 5). Verifier 1 is robust to embedder choice: OpenAI text-embedding-3-large yields a strict subset of mxbai’s candidates at every threshold ($\rho=0.78$; Appendix A). For Verifier 3, a GPT-4o swap on a 50-problem subset gives $\kappa=0.44$, and a full local open-weight re-score (Qwen3-32B, identical released templates) spans κ 0.24–0.82 across all re-derivable cells (Appendix D, Table 4). The local judge is more lenient than Sonnet in every re-derivable cell, and the GPT-4o swap is likewise more lenient on its subset, so the reported absolute numbers sit at the conservative edge of the judge envelope. The leniency margin is model-dependent (up to +37 pp on baselines), so cross-model gaps can reverse: seed 42 leads Qwen3-VL-32B on PhysReason under Sonnet but trails it under the local judge (Finding 5). The headline claim survives the swap. Under the local judge the seed-42 checkpoint scores 51.9 on PhysReason and 52.5 on PUB-OE, against the base’s 49.1 and 52.3. On OlympiadBench-Physics it scores 52.2 against the base’s 40.8, and on PHYSOLYM-A 40.2 against 27.0: every lift keeps its sign. The 52.2 value comes from responses regenerated greedily from the released checkpoint (originals unpublished; leniency-band context in Appendix D).

Limitations. Five limitations bound the conclusions. (i) *LLM-judged scoring.* Per Finding 5, absolute open-ended scores are judge-dependent; we mitigate by fixing and releasing the judge, publishing every per-record verdict, and reporting judge-free strict scores. (ii) *Cross-lingual scope.* The translation finding is Sonnet-4.5-specific, on 59 paired items (Appendix H.4). (iii) *Topic coverage.* The corpus and evaluation skew toward competition-canonical areas (per-category accuracy in Appendix H.1); experimental methods, condensed matter, and plasma physics are underrepresented. (iv) *Construct validity.* Olympiad problems probe closed-form competition reasoning, not experimental-design or open-ended modeling skill. (v) *Difficulty calibration.* Difficulty labels inherit the organizers’ student-targeted calibration.

Conclusion. In sum: five findings document where a physics training-and-evaluation system silently breaks. Three released verifiers, the artifacts they certify, and a recipe that lifts the base 18.3 pp close the gaps; every measurement ships publicly for re-verification.

Acknowledgments

The author thanks Kevin Zhou for granting permission to redistribute his olympiad handouts under CC BY-NC 4.0, the Estonian Physics Olympiad committee for making their archived problems and solutions publicly available at <https://fyysika.ee/>, the international olympiad committees (IPhO, NBPhO, EuPhO, APhO, USAPhO, INPhO) for the public archives that enabled the novel-source held-out evaluation, and the maintainers of the public physics-VL benchmarks (PhyX, MMMU-Pro, OlympiadBench, UGPhysics, PhysReason, PhysUniBench) whose released training pools and evals enabled the contamination audit reported in this work. Compute support for Physics-R1 training and evaluation was provided by RunPod.

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Algorithm 1 Three-stage contamination audit.

Require: Train pool T , external corpora $\{E_k\}_{k=1}^K$, held-out splits $\{H_j\}_{j=1}^J$, normalize fn $\text{norm}(\cdot)$, embedder $\text{enc}(\cdot)$, LLM judge $\text{JUDGE}(\cdot, \cdot) \in \{\text{close-dup}, \text{topic-neighbor}\}$, thresholds $\tau_J=0.4$, $\tau_C=0.85$

Ensure: Audited pool T' disjoint from $\bigcup_k E_k \cup \bigcup_j H_j$ at the joint thresholds.

Stage 1: 5-gram Jaccard (n-gram audit).
1: $S_t \leftarrow \{5\text{-gram shingle set of } \text{norm}(t)\}$ for each $t \in T \cup \bigcup_k E_k \cup \bigcup_j H_j$
2: **for** $t \in T$ **do**
3: $J_{\max}(t) \leftarrow \max_{x \in \bigcup_k E_k \cup \bigcup_j H_j} |S_t \cap S_x| / |S_t \cup S_x|$
4: **end for**
Stage 2: $\text{max-ai-embed-large cosine}$ (paraphrase recall).
5: $\mathbf{e}_t \leftarrow \text{enc}(\text{norm}(t)) / \|\text{enc}(\text{norm}(t))\|$ for each t
6: **for** $t \in T$ **do**
7: $C_{\max}(t) \leftarrow \max_{x \in \bigcup_k E_k \cup \bigcup_j H_j} \mathbf{e}_t^\top \mathbf{e}_x$
8: **end for**
9: $C(T) \leftarrow \{t : J_{\max}(t) \geq \tau_J \text{ OR } C_{\max}(t) \geq \tau_C\}$ ▷ candidate set, high-recall union
Stage 3: Haiku-4.5 LLM-judge (precision filter). For each $t \in C(T)$ with top-matching $x^*(t) \leftarrow \arg \max_x \mathbf{e}_t^\top \mathbf{e}_x$, query JUDGE to classify the pair as a close duplicate (paraphrase / numeric variation of the same problem) or a same-topic neighbor (related physics, distinct setup).
10: $R(T) \leftarrow \{t \in C(T) : \text{JUDGE}(t, x^*(t)) = \text{close-dup}\}$
11: $T' \leftarrow T \setminus R(T)$
12: **return** T' and per-stage counts $|J_{\max} \geq \tau_J|, |C_{\max} \geq \tau_C|, |R(T)|$ (Tables 1, 3).

A Audit pipeline details and worked examples

Stage-3 LLM-judge: SciInstruct cosine-bucket near-duplicate rate. For each of the 4,846 SciInstruct $\leftrightarrow$ eval Stage-2 candidate pairs ($\cos \geq 0.85$), the Stage-3 Haiku-4.5 judge receives both problem statements and returns *close duplicate* (paraphrase / numeric variation of the same problem) or *same-topic neighbor* (related physics, distinct setup). The close-duplicate share is sharply threshold-driven: at $\cos \geq 0.95$ every flagged pair is a close duplicate; at the threshold edge $[0.85, 0.87]$ only 1.5% are. Stage-2 still surfaces genuinely close physics content at the low-cos end—the topic overlap is real, just not strict duplication.

Cosine bucket	N pairs	close duplicate	same-topic neighbor	% close-dup
$[0.95, 0.99)$	17	17	0	100.0%
$[0.90, 0.95)$	127	10	117	7.9%
$[0.87, 0.90)$	1,159	54	1,105	4.7%
$[0.85, 0.87)$	3,543	53	3,490	1.5%
Total	4,846	134	4,712	2.8%

The pattern matches the threshold-sensitivity table (Table 3): $\cos \geq 0.95$ is precision-dominant, ≥ 0.85 is recall-dominant; Stage-3 is the precision filter that converts a recall-dominant Stage-2 candidate set into a high-precision near-duplicate set. Per-eval Stage-3 near-duplicate counts (out of Stage-2 raw, matching Table 1): PhysReason-full 36/2,687 (1.3%), PhysUniBench-en 22/1,027 (2.1%), PhyX-mini 46/703 (6.5%), OlymBench-Phys 15/130 (11.5%), PHYSOLYM-A 8/163 (4.9%), MMMU-Pro 7/141 (5.0%).

External-corpora $\leftrightarrow$ held-out pairwise audit (Stage 2, OlympiadBench shared-source). The Stage-1 / Stage-2 cells of the main contamination matrix (Table 1) cover competitor training pools against the held-out evals. The complementary cross-channel audit below pairs the four public physics-olympiad benchmarks against PHYSOLYM-A and PhyX 1000q, exposing shared-source paraphrase overlap between Olympiad-style problems composed independently. The single Stage-1 hit *OlympiadBench-Physics* $\rightarrow$ PHYSOLYM-A is the EuPhO 2020 “Mechanical accelerator” problem that grounds the 99.8% (rather than 100%) novel-source claim.

	PHYSOLYM-A	PhyX 1000q
OlympiadBench-Physics (692)	1 / 136	0 / 2
PhysReason-mini (200)	0 / 2	—
PhysReason-full (1,200)	0 / 35	—
PhysUniBench-en (1,022)	0 / 27	—

Stage 1 (5-gram Jaccard). Tokenize each problem statement with a unicode word tokenizer; build the 5-gram shingle set; compute Jaccard similarity over shingle sets; flag pairs with similarity ≥ 0.4 . The threshold is calibrated against worked examples: an OpenStax College Physics $\leftrightarrow$ University Physics duplicate scores Jaccard 1.0; a UGPhysics paraphrase missed by Stage 1 (Jaccard 0.31) is flagged at Stage 2 (cosine 0.91); plausible misses include numerical-value substitution (same setup, different constants) and translation across languages. Stage 2 (embedding cosine). Encode each problem statement with `mxbai-embed-large-v1` [Lee et al., 2024] (1024-dim normalized embeddings); compute pairwise cosine; flag pairs with cosine ≥ 0.85 .

A.1 Threshold-Sensitive Leakage Finding on a Researcher-Curated Baseline

Table 3: **Threshold-sensitivity of the leakage finding on a researcher-curated baseline.** A 1,679-record sample drawn from PHYSCORP-PRE-AUDIT (the released 14,294-record pre-audit pool) under conventional 5-gram-Jaccard + within-pool embedding dedup is paired against a 500-record internal analysis eval; each cell reports records flagged at $J \geq j_{\text{thr}}$ or $\cos \geq c_{\text{thr}}$. The boxed cell ($J \geq 0.4$, $\cos \geq 0.85$) is the operating point referenced throughout the paper. Stage-1 alone (Jaccard) is bimodal: all 56 leaks are exact at $J=1.0$. Stage-2 (cosine) catches an additional ~ 92 paraphrases the n -gram audit misses at the published threshold; loosening cosine to 0.80 pushes the leak rate above 27%.

cosine threshold	$J \geq 0.3$	$J \geq 0.4$	$J \geq 0.5$
$\cos \geq 0.80$ (lax)	455 (27.1%)	455 (27.1%)	455 (27.1%)
$\cos \geq 0.85$ (paper op.)	148 (8.8%)	148 (8.8%)	148 (8.8%)
$\cos \geq 0.90$ (strict)	79 (4.7%)	79 (4.7%)	79 (4.7%)
<i>Single-stage rates (no union):</i>			
Stage-1 only ($J \geq j_{\text{thr}}$, no $\cos$)	56 (3.3%)	56 (3.3%)	56 (3.3%)
Stage-2 only ($\cos \geq c_{\text{thr}}$, no J)	455 (27.1%)	148 (8.8%)	79 (4.7%)

To ground Finding 1 we audit a researcher-curated baseline—a 1,679-record sample drawn from PHYSCORP-PRE-AUDIT (the released 14,294-record pre-audit pool) under conventional 5-gram-Jaccard + within-pool embedding dedup, paired against a 500-record internal analysis eval. The 500-record eval is held internal to ground this finding and is distinct from PHYSOLYM-A (constructed post-audit); the 1,679-record sample is reproducible from the released PHYSCORP-PRE-AUDIT so the audit can be re-run by downstream users. Stage-1 catches 56 records (3.3%) all at $J=1.0$ (bimodal distribution—verbatim duplication under our normalization). Stage-2 cosine ≥ 0.85 flags an additional 92 records (5.5%), bringing the joint leak rate to 148 (8.8%); the cosine dimension sweeps 4.7–27.1% across $\cos \in \{0.90, 0.85, 0.80\}$ while Jaccard is flat at 3.3% (Table 3). The 148 flagged decompose into 56 exact ($J=1.0$), 23 strong paraphrases ($\cos \geq 0.90$), 69 weak paraphrases ($0.85 \leq \cos < 0.90$). The 5.5-pp gap is the rephrasing dark-matter that single-stage audits miss because the *same* problem appears across upstream aggregations under wording that evades $J \geq 0.4$ but trips cosine ≥ 0.85 .

Pipeline reproducibility note. Threshold-sensitivity numbers were produced by `audit_pipeline/threshold_sensitivity.py`. Normalization: lower-case, strip LaTeX commands (`\frac`, `\sqrt`, ...), remove `\{``}``[``]``(``)` delimiters, collapse whitespace, drop  $< 5$ -word shingles. Embedding: `sentence-transformers 5.4.1`, batch 32,  $L_2$ -normalized. Best-overlap via inverted-index pruning (Stage 1) and full  $N \times M$  matmul (Stage 2). Saved scores in `threshold_sensitivity_scores.npz`.

Bimodal Jaccard distribution. Every Stage-1 leak in our researcher-curated baseline sits at $J=1.0$: aggressive normalization collapses surface variance, so shared problem statements yield

identical shingle sets while distinct records land below $J=0.3$. Cosine ≥ 0.85 is therefore the sole paraphrase-class detector here; bimodality is corpus-specific.

Re-audit and the cleaned PhysR1Corp. Starting from a 2,433-record candidate closed-form pool, three sequential cleanup passes against the six paper-canonical comparison corpora (PhyX, MMMU-Pro Physics, OlympiadBench-Physics, PhysReason-full, PhysUniBench-en, PHYSOLYM-A) produced the released PHYSR1CORP: (i) MMMU-Pro Physics joint $J \geq 0.4 \vee \cos \geq 0.85$ re-audit dropped 87 records (Stage-1: 16 records, $16/60=26.7\%$ MMMU-Pro coverage; Stage-2 union: 87 records, $52/60=86.7\%$); (ii) PhyX-mini cosine ≥ 0.85 Stage-2 audit dropped 69 records, all real-near-duplicate physics-problem variants confirmed by manual inspection (e.g. MgF_2 anti-reflection-coating problem with slightly tweaked n_{glass} and option labels, top cos 0.93); (iii) PhysUniBench-en cosine ≥ 0.85 Stage-2 audit dropped 9 template duplicates (top cos 0.93, shared problem-template stems). Total dropped: $87 + 69 + 9 = 165$ records (with 3 records flagged in multiple channels netting to $87 + 78 = 165$ unique). The released **2,268**-record PHYSR1CORP retains 19 Stage-2 hits across PhysOlym-A (3), MMMU-Pro (1), OlymBench-Phys (4), PhysReason-full (11) classified as same-topic neighbors on manual inspection (top cos ≤ 0.87 , different problem setups, dimensionalities, or geometries; Table 1). MMMU-Pro Physics remains excluded from the headline Table 2, the saturation-gradient narrative in §5.2, and the dense-vs-binary ablation in §5.3, because the eval is small (60 records) and prior work has flagged it as contaminated against multiple frontier training corpora; Sonnet’s MMMU-Pro Physics number is reported only as a frontier-model reference (Sonnet was not trained on PHYSR1CORP).

Embedding-model and threshold rationale. We chose mxbai-embed-large-v1 for Stage 2 over bge-large-en-v1.5, e5-large-v2, OpenAI text-embedding-3-large, and Voyage voyage-3 because it (i) is permissively licensed (Apache 2.0, no API dependency); (ii) ships 1024-dim normalized vectors with cosine tuning; (iii) scored highest on MTEB physics-adjacent retrieval at audit time. The $\cos \geq 0.85$ threshold was calibrated against worked examples (UGPhysics paraphrase: $J=0.31$ Stage-1-miss, $\cos=0.91$ Stage-2-catch); the $\cos \in \{0.80, 0.85, 0.90\}$ sensitivity grid brackets the operating point.

Embedder-sensitivity ablation (mxbai vs. text-embedding-3-large). We re-encode the released PHYSCORP-PRE-AUDIT pool (14,294 records) and PHYSOLYM-A (500 records) under OpenAI text-embedding-3-large and compute pairwise cosines, then compare per-train-record max-cosine rankings against the mxbai baseline. **Spearman** $\rho=0.78$ ($p \approx 0$); the candidate-set relationship at the operating threshold is summarized below.

Threshold cos $\geq$	mxbai cand.	text-embedding-3-large cand.	both	only mxbai	only OpenAI
0.85 (paper op.)	763	514	514	249	0
0.87	631	512	512	119	0
0.90	551	511	511	40	0
0.95	520	506	506	14	0

text-embedding-3-large flags a *strict subset* of mxbai’s candidates at every threshold (only-OpenAI count is 0 at all four levels): every record text-embedding-3-large would catch is also caught by mxbai. mxbai is therefore the more conservative (higher-recall) Stage-2 embedder, and the audit cannot have missed any contamination that text-embedding-3-large would have surfaced. The candidate-set Jaccard at the operating threshold is 0.67. *Caveat.* This ablation is on the user-facing audit case (released pool $\leftrightarrow$ released eval), not the SciInstruct competitor-pool case from Table 1; the strict-subset direction does not formally transfer, but the $\rho=0.78$ rank correlation suggests the SciInstruct 134-near-duplicate count is robust in direction under embedder change. Sensitivity ablation against voyage-3 is left to follow-up work.

Stage-3 judge model dependence and reproducibility. Stage-3 introduces a dependence on Anthropic’s Haiku-4.5 (the close-duplicate vs. same-topic-neighbor classifier). To ensure long-term reproducibility we (i) pin the exact model identifier (claude-haiku-4-5 as of 2026-05) in the released audit pipeline (Stages 1–2 as `audit_two_stage.py`; the Stage-3 prompt template and pinned model id ship alongside); (ii) release the full per-pair judge prompts and per-pair verdict labels (`judge_label` arrays) alongside the cosine scores in

threshold_sensitivity_scores.npz, so the contamination flag set is reproducible without re-querying the API; (iii) document a fallback protocol (Sonnet 4.5 or GPT-4o on the same prompt template) for cases where Haiku-4.5 access is no longer available. Because Stage-3 is a precision filter applied only to Stage-2 candidates ($\leq 0.5\%$ of the train pool), its labels are also the most amenable to manual re-verification by a downstream auditor; the cosine-bucketed precision profile (Table A) gives a cheap calibration signal against any future judge.

B Held-out splits and corpus annotation schema

The released splits are PHYSCORP-A (the 6,432-record audited corpus) with its closed-form RL carve-out PHYSR1CORP (2,268 records) and PHYSOLYM-A (the 500-problem held-out olympiad eval). The annotation schema below applies uniformly across all three.

Eight-field annotation schema. Each record carries the following annotations, generated by Sonnet 4.5 batch annotation (3,900 records with full annotation; the remainder carry source-native labels merged into the same schema for $\sim 31,000$ total label values):

- `difficulty` $\in \{1,2,3,4,5\}$ (Sonnet-aggregated), plus optional source-native: Estonian organizer-issued 1–10, Zhou pedagogical 1–5, Zhou advanced [A] flag.
- `concept` $\in \{\text{Mechanics, Electromagnetism, Quantum, Thermodynamics, Waves, Optics, Modern, Relativity, Particle}\}$.
- `problem_type` $\in \{\text{Conceptual, Computational, Proof-based, Experimental}\}$.
- `expected_solution_length` $\in \{\text{S, M, L}\}$ (short / medium / long).
- `math_level` $\in \{\text{Algebra, Calculus, Vector, LinearAlgebra, DiffEq}\}$.
- `modality` $\in \{\text{text, multimodal}\}$.
- `language` (BCP-47): en, et, en-et (bilingual paired).
- `license` (SPDX-style): CC-BY-4.0, CC-BY-SA-4.0, CC-BY-NC-4.0, Public-Domain.

Worked example record (JSONL). A typical record from PHYSOLYM-A:

```
{
  "index": "estonian_2017_lahtine_2",
  "source": "estonian_olympiad",
  "license": "CC-BY-NC-4.0",
  "language": "en-et",
  "messages": [{"role": "user", "content": "An ideal gas..."}],
  "solution": "By the first law...\boxed{1.5}",
  "images": [],
  "concept": "Thermodynamics",
  "difficulty": 4,
  "native_difficulty": {"scale": "1-10", "value": 7},
  "problem_type": "Computational",
  "expected_solution_length": "M",
  "math_level": "Calculus",
  "modality": "text",
  "audit_passed": true
}
```

Inter-annotator agreement. For the 3,900-record fully-annotated subset, Sonnet 4.5 is run with two seeds on the same 100 records (random sample, seed 42); per-field Cohen’s κ between the two annotation runs is reported in the released dataset card. Preliminary inspection shows $\kappa \geq 0.85$ on concept and problem_type (categorical with sharp boundaries), $\kappa \sim 0.7$ on difficulty (ordinal with neighboring-class confusion), and $\kappa \sim 0.6$ on expected_solution_length. The lower κ on expected_solution_length reflects genuine ambiguity in the medium-vs-long boundary; downstream users requiring stable solution-length labels should treat the field as a noisy proxy.

Native difficulty preserved. For the 27% of PHYSOLYM-A records with Estonian native difficulty 1–10 and the 38% with Zhou pedagogical 1–5 point values, the *Sonnet-aggregated* difficulty field is reported for cross-source consistency, but the *native* difficulty is preserved in a separate `native_difficulty` field with explicit `scale` and `value` keys. The Sonnet difficulty curve in Table 13 uses native Estonian 1–10, not the aggregated 1–5, because native labels avoid the self-annotation circularity in which the difficulty estimate depends on the same model whose accuracy is being measured.

C Reward function and full hyperparameter table

This appendix specifies both reward shapes referenced from §4: the recommended binary correctness reward r_{bin} defined inline in §4 (§C.1) and the dense five-component reward of Equation 2 reported as an ablation (§C.2). Both share the matching/extraction primitives below; binary uses r_{ans} alone, dense composes all five components and clips.

C.1 Binary correctness reward (recommended)

The recommended reward is

$$r_{\text{bin}}(y, x) = \mathbb{I}[\text{MATCH}(\text{EXTRACTBOXED}(y), g(x))] \in \{0, 1\},$$

where `EXTRACTBOXED` parses the last `\boxed{}` via brace-counting (handles unlimited nesting, e.g. `\sqrt{\frac{T_0}{\eta}}`) and `MATCH` accepts: (i) MCQ-letter equality on `{A,B,C,D,...}` gold; (ii) multi-part numeric agreement within $\pm 1\%$ relative tolerance, after `latex_to_plain` normalization (`\text{\}`/`\mathrm{\}` stripped, `\frac{a}{b}` $\rightarrow$ `(a)/(b)`, `\times/\cdot` $\rightarrow$ `*`, `\pi` $\rightarrow$ `pi`), `float()`, then `eval()` on expression-like strings, then prefix-numeric extraction; (iii) symbolic equivalence via `sympy.simplify(expr_pred - expr_gold) == 0` for symbolic gold. Released as `reward_physics.py`; selected by env var `DENSE_REWARD=0` (the default), which returns r_{ans} as a clean 0/1 binary.

Why binary is the deployable default (full theoretical analysis). The (P1)/(P2) intuitions are stated inline in §4; we add the (P3) derivation here. (P1) Group-normalization renders A_k invariant to affine rescaling of r within a group, so dense shaping only matters when it *reorders* rollouts; we measure 14.3% of within-group pairs flipped by dense, 87% inside the all-wrong subgroup. (P2) Those wrong-group flips reward LaTeX-format proxies satisfiable without solving the physics—a Goodhart channel that hurts prose-and-equation open-ended evaluation; the matched-step-60 binary-vs-dense gap is largest on the audited PHYSOLYM-A-liberal split.

(P3) Binary reward maximizes per-prompt advantage variance after the difficulty curriculum.

The MM-Eureka curriculum drops prompts where all K rollouts are correct or all are wrong, so on every surviving prompt the rollout-correct rate is $p \in (0, 1)$. Binary reward is then a Bernoulli over rollouts: $\bar{r} = p$, $\sigma_r^2 = p(1 - p)$, and the group-normalized advantages take exactly two values

$$A_k^{\text{bin}} = \begin{cases} \sqrt{(1-p)/p} & r_k = 1 \\ -\sqrt{p/(1-p)} & r_k = 0 \end{cases}, \quad \text{Var}(A^{\text{bin}}) = 1. \quad (3)$$

The advantage variance is exactly 1, the maximum a K -sample group-normalized estimator can carry on a Bernoulli reward. Adding a bounded shaping term $\delta_k \in [0, \Delta]$ with $\Delta=0.45$ (the $r_{\text{fmt}} + r_{\text{dim}} + r_{\text{sym}}$ budget of the dense reward) inflates the within-group standard deviation σ_r by an $O(\Delta^2)$ term while leaving the between-correctness mean separation roughly unchanged, so the magnitude of the average correct-vs-wrong advantage *shrinks*:

$$|A_{\text{correct}}^{\text{dense}}| \approx \frac{1-p}{\sqrt{p(1-p) + \Delta^2/12}} < \frac{1-p}{\sqrt{p(1-p)}} = |A_{\text{correct}}^{\text{bin}}|. \quad (4)$$

Dense reward thus trades *between*-correctness gradient signal-to-noise for *within*-correctness rank information, but the within-correctness flips are exactly the Goodhart channel of (P2).

C.2 Dense five-component physics-native reward (ablation)

The dense five-component Physics-R1 reward (Equation 2, Section 4) is implemented as $r = r_{\text{ans}} + r_{\text{fmt}} + r_{\text{dim}} + r_{\text{sym}} + r_{\text{cons}}$, clipped to $[-1, 1]$.

Per-component implementation. Full code in `reward_physics.py`. $r_{\text{ans}} \in \{0, +1\}$: brace-counting `\boxed` parser; MCQ-letter equality; numeric/symbolic via `latex_to_plain` normalization (`\frac`, `\text`, `\pi`, `\times/\cdot`) then `float/eval/prefix-numeric`; multi-part requires all parts $\pm 1\%$. $r_{\text{fmt}} \in \{0, +0.1\}$: non-empty `\boxed{\}`. $r_{\text{dim}} \in \{0, +0.15\}$: number-prefix-guarded regex extracts unit tokens, mapped to `sympy.physics.units` (32 tokens); fired only when every detected unit resolves. $r_{\text{sym}} \in \{0, +0.20\}$: first-success `sympy.simplify` on LaTeX-cleaned `\frac{NUM}{DEN}` intermediates. $r_{\text{cons}} \in \{-0.25, 0\}$: negative-only penalty when energy/momentum balance from corpus annotation is violated by $> 5\%$ relative.

Mode and version pins. Released as `reward_physics.py`. The mode is selected by env var `DENSE_REWARD`: 0 (default, recommended) returns r_{ans} only as 0/1 binary, recovering r_{bin} of §C.1; 1 returns the full clipped dense sum (ablation). The audit factor (env `AUDIT_LAMBDA`) zeros out reward on contaminated training items when set. Version pins: `sympy == 1.13.3`, `transformers == 4.57.0`, `vllm == 0.11.0`, `verl == 0.6.1`.

Worked example (rollout group; binary vs. dense advantages). A concrete realization of the (P1)/(P2) Goodhart channel of §4 on one prompt with $K=8$ rollouts. The prompt is a two-step kinematics problem with gold `19.6` N (problem id `efo-2014-3a`); rollouts $y_1, \dots, y_4$ commit to the correct answer with varying CoT quality, $y_5, \dots, y_8$ commit to wrong answers ranging from arithmetic-slip (`19.8`, $>1\%$) to off-by-physics (`42.0`).

k	Final <code>[·]</code>	CoT shape	r_{ans}	r_{fmt}	r_{dim}	r_{sym}	r_{cons}	r_{bin}	r_{dense}
1	19.6	full units + <code>\frac</code>	1	0.10	0.15	0.20	0	1.00	1.00
2	19.6	units only, no <code>\frac</code>	1	0.10	0.15	0	0	1.00	1.00
3	19.6	<code>\frac</code> only, no units	1	0.10	0	0.20	0	1.00	1.00
4	19.6	sparse CoT (“answer is 19.6”)	1	0.10	0	0	0	1.00	1.00
5	42.0	full units + <code>\frac</code>	0	0.10	0.15	0.20	0	0.00	0.45
6	19.8	units only, no <code>\frac</code>	0	0.10	0.15	0	0	0.00	0.25
7	42.0	<code>\frac</code> only, no units	0	0.10	0	0.20	0	0.00	0.30
8	no <code>[·]</code>	rambling, no commit	0	0	0	0	0	0.00	0.00

After clipping ($r \leq 1$) the four correct rollouts collapse to identical reward under both shapes. After group-normalization (Eq. 1) the binary advantage vector is $A^{\text{bin}} = (+1, +1, +1, +1, -1, -1, -1, -1)$, every correct rollout receives equal positive gradient and every wrong rollout equal negative gradient. The dense advantage vector is $A^{\text{dense}} \approx (+1.04, +1.04, +1.04, +1.04, -0.55, -1.00, -0.93, -1.69)$ (computed exactly: $\bar{r} = 0.625$, $\sigma_r \approx 0.428$). *Three observations land the (P1)–(P3) theory at the sample level:*

- (P1) realized. Among the four correct rollouts ($k=1, 2, 3, 4$), dense and binary assign *the same* advantage to every rollout—rank-equivalent in the correct subgroup, even though rollouts 1–3 “deserve” more credit by the *a priori* physics-native intuition. Clipping at 1 erases the dense-side variation among correct rollouts.
- (P2) realized. Among the four wrong rollouts ($k=5, 6, 7, 8$), dense reorders them by LaTeX surface form: $k=5$ (well-formatted, units, `\frac`, `42.0` way wrong) gets the *smallest* negative advantage (-0.55), $k=8$ (no boxed commit) gets the most negative (-1.69). The policy gradient is therefore pushed toward producing well-formatted wrong reasoning over poorly-formatted wrong reasoning—the canonical Goodhart channel. Note $k=5$ has dense advantage -0.55 vs. binary -1.00 : the wrong-answer gradient is *weakened* for the format-compliant rollout, exactly the bias toward proxy satisfaction.
- (P3) realized. The magnitude of the average correct-rollout advantage is $|A_{\text{correct}}^{\text{bin}}| = 1.0$ vs. $|A_{\text{correct}}^{\text{dense}}| \approx 1.04$ (very close because clipping caps both at $r=1$). The magnitude of the average wrong-rollout advantage is $|A_{\text{wrong}}^{\text{bin}}| = 1.0$ vs. $|A_{\text{wrong}}^{\text{dense}}| \approx 1.04$ on the dense side as well, but the within-wrong spread of $\sigma = 0.42$ across $\{-0.55, -0.93, -1.00, -1.69\}$ is the within-group rank-flipping variance that absorbs gradient capacity into the Goodhart direction. Binary spends zero capacity on within-correctness ranking and all of it on the correct-vs-wrong axis.

The aggregate effect of running this calculus across 60 RL steps of 96-prompt batches over the full 2,268-prompt PHYSR1CORP pool (≈ 2.5 epochs) is the matched-step-60 binary-vs-dense gap of Table 2 (problem-level liberal Sonnet-judge accuracy across all open-ended columns;): PhysReason 32.2 vs. 23.3 (+8.9 pp), PUB-OE 37.0 vs. 37.7 (−0.7 pp, tied), OlymBench-Phys liberal 45.4 vs. 40.5 (+4.9 pp), PHYSOLYM-A liberal 25.6 vs. 19.2 (+6.4 pp).

Hyperparameter and framework details. The full GSPO+DAPO configuration with all flags, lambdas, clip ranges, dynamic-sampling settings, and the difficulty-curriculum thresholds is in Table 11 (Appendix H.7) and the released YAML. The ver1 0.6.1 FSDP1 reproducibility note (Appendix F) and the upstream GitHub issue link are tracked in the released README.

Algorithm 2 Dense five-component physics-native reward for one rollout.

Require: Solution string s , gold answer g , optional conservation flag cons from `extra_info`
Ensure: $r \in [-1, 1]$

- 1: $\text{ans} \leftarrow \text{EXTRACTBOXED}(s)$ ▷ brace-counting parser; nested \boxed OK
- 2: $r_{\text{ans}} \leftarrow +1$ if $\text{MATCH}(\text{ans}, g)$ under MCQ-letter / multi-part / $\pm 1\%$ numeric tolerance, else 0
- 3: $r_{\text{fmt}} \leftarrow +0.1$ if ans is non-empty (well-formed `\boxed{\dots}`), else 0
- 4: $U \leftarrow \text{EXTRACTUNITS}(s)$ ▷ regex (number)(unit)(exponent) with number-prefix guard
- 5: $r_{\text{dim}} \leftarrow +0.15$ if $|U| \geq 1$ and $\forall u \in U : \text{RESOLVESINSYMPY}(u)$, else 0
- 6: $F \leftarrow \text{EXTRACTFRACS}(s)$ ▷ find all `\frac{NUM}{DEN}`
- 7: $r_{\text{sym}} \leftarrow +0.20$ if $\exists (\text{NUM}, \text{DEN}) \in F : \text{SYMPIFIES}(\text{NUM}) \wedge \text{SYMPIFIES}(\text{DEN})$, else 0
- 8: **if** cons is provided ($\sim 3,100$ -record subset) **then**
- 9: $\hat{p} \leftarrow \text{TRYFLOAT}(\text{ans})$; $p^* \leftarrow \sum \text{cons.in} - \sum \text{cons.out} \setminus \text{ans}$
- 10: $r_{\text{cons}} \leftarrow -0.25$ if $|\hat{p} - p^*| / \max(|p^*|, |\hat{p}|, \varepsilon) > 0.05$, else 0
- 11: **else**
- 12: $r_{\text{cons}} \leftarrow 0$ ▷ negative-only; no positive reward for satisfaction
- 13: **end if**
- 14: **return** $\text{clip}(r_{\text{ans}} + r_{\text{fmt}} + r_{\text{dim}} + r_{\text{sym}} + r_{\text{cons}}, -1, 1)$

D Verifier 3 details: the three judge scripts of the liberal layer, scoring conventions, and reproducibility

This appendix documents the three Sonnet-4.5-as-judge variants used in Table 2 and the scoring conventions adopted across columns. Source code for all three judges is released in the `judge/` directory of the code repository (github.com/shanyang-me/physics-r1-neurips2026).

Why three judges. Open-ended physics olympiad problems differ structurally: PhysReason and PhysUniBench-OE problems are explicitly multi-sub-part (often 2–5 sub-questions per record, each with its own gold answer), while PHYSOLYM-A and OlympiadBench-Physics problems are graded at the problem level (a single gold solution document, with the model’s final answer compared against it). A single judge prompt cannot serve both. We therefore use three judges, each tuned to its eval’s structure:

- `judge_olympiad.py` (**problem-level**, used for PHYSOLYM-A and OlympiadBench-Physics). One Sonnet 4.5 call per problem; the prompt provides the full gold solution paragraph and a `\boxed{\dots}`-extracted candidate answer (with a 600-char response-tail fallback if no boxed is emitted), and asks for a single YES/NO verdict on whether the candidate’s final answer is mathematically/physically equivalent to the gold’s. Tolerance is 2% relative.
- `llm_judge_v2_alignment.py` (**per-subpart, AND across sub-parts**, used for PhysReason). For each gold sub-answer g_i , a separate Sonnet 4.5 call asks: *does ANY of the candidate’s predictions equal g_i ?* Per-subpart verdict is YES/NO. `judge_problem_correct` is the AND across all sub-parts. Tolerance: 1%.
- `llm_judge_v3_pubeo.py` (**per-subpart, AND across sub-parts, with cached clean gold + tail fallback**, used for PhysUniBench-OE). Same per-subpart structure as v2, but

with a pre-extracted *clean* per-subpart gold list (e.g., ["2.68nC", "7853.1W"]) that bypasses PUB-OE’s verbose paragraph-form gold. If the candidate’s `\boxed{}` list is empty, a regex fallback scans the last 300 chars of the response for likely numeric/symbolic answers. Per-subpart verdict is YES/NO; `judge_problem_correct_v3` is the AND across all sub-parts. Tolerance: 2%.

Scoring conventions in Table 2. All open-ended cells use *problem-level* accuracy: for multi-sub-part problems (PhysReason, PhysUniBench-OE) every sub-part must be judged correct for the problem to count, via the `judge_problem_correct` field of the v2/v3 judges (AND across sub-parts); for problem-level evals (PHYSOLYM-A, OlympiadBench-Physics) `judge_olympiad.py` returns one YES/NO per problem. We also computed a softer *per-subpart* variant (partial credit, $\sum_i \mathbb{1}[\text{sub}_i \text{ correct}] / \sum_i 1$) for the multi-sub-part columns and verified it is uniformly 4–17 pp higher across rows; per-subpart values are released alongside the dataset for users who want to study the partial-credit lens but are not reported as headline in Table 2.

Reproducibility note. All Sonnet-judge runs in Table 2 use `workers=2–4` concurrency; errored sub-judgments are filtered and re-judged at lower concurrency rather than counted as wrong. Per-cell judge-error counts (typically $\leq 1\%$ of records) and per-record verdicts are released in the supplementary archive at `judge_audit.json`, so downstream auditors can verify each cell independently. The Sonnet 4.5 PhysReason responses (Table 2, [†]) are regenerated at the 16,384-token response budget used by all open-source baselines: intermediate-length Sonnet responses under default API settings (mean under 200 characters) systematically fail to commit a boxed final answer on multi-part problems, which the v2 judge scores as wrong.

Verbatim judge prompt (problem-level, PHYSOLYM-A + OlymBench). The Sonnet 4.5 judge is invoked with the following template:

```
You are grading a physics olympiad answer.

GOLD (full reference solution; the final numeric/symbolic answer
is what matters):
{gold}

CANDIDATE answers (extracted from the model’s \boxed{} markers):
{preds}

(If the candidate emitted no \boxed{}, the candidate is the last
600 chars of its full response:)
{tail}

Task: decide whether the candidate’s final answer is
mathematically/physically equivalent to the gold’s final answer.
Allow: different but equivalent algebraic forms; trivial
unit/format differences; rounding within 2% relative tolerance;
trailing prose.
Reject: different magnitude, different sign, different functional
form, missing or wrong physical content, no answer.

Respond with EXACTLY one word: YES or NO.
```

Verbatim judge prompt (per-subpart, PhysReason + PhysUniBench-OE). For each gold sub-answer g_i , the judge is invoked with:

```
You are grading a physics olympiad answer.

GOLD answer: {g_i}

CANDIDATE predictions (one or more, separated by ==):
{preds}

Task: decide whether ANY of the candidate predictions is
mathematically/physically equivalent to the gold answer.
```

Allow: different but equivalent algebraic forms; trivial unit/format differences ("450 N" == "450 \text{ N}"); rounding within 1-2% relative tolerance; trailing prose ("approximately", "to the right"); different variable names mapping cleanly.
 Reject: different magnitude, different sign, different functional form, missing or wrong physical content.

Respond with EXACTLY one word: YES (if any candidate matches) or NO. No other text.

Per-seed reproducibility notes. Two policy choices in the released per-seed exports deserve disclosure. First, the seed-42 PhysReason export contains 1,258 rows for 1,200 problem ids (58 duplicates); the published 32.2% (and hence the 39.6 ± 6.4 mean) is computed over all rows, and deduplication would shift that seed to 32.9–33.8%. Second, the seed-17 and seed-23 PHYSOLYM-A exports use a scored-judge format in which unjudgeable and partial rows (43% and 39% of records respectively) are counted as incorrect, so the published 25.0 and 28.2 are lower bounds under that policy; the seed-42 export uses the binary YES/NO judge on all 500 records, with 13 judge-failure rows defaulted to incorrect. All per-record verdicts behind these numbers are public.

Local open-weight judge re-score. We re-scored every re-derivable LLM-judged column with Qwen3-32B deployed locally (vLLM, greedy, thinking disabled), using the verbatim prompt templates of the released judge scripts and the same policies (unparseable verdicts count as incorrect; deterministic strict-norm sub-verdicts in the v3 protocol are kept fixed and only LLM-judged sub-verdicts are re-judged). Table 4 reports problem-level agreement with the pinned Sonnet verdicts. The seed-17/23 PHYSOLYM-A exports store verdicts without the underlying responses, so they cannot be re-judged from published data; OlympiadBench-Physics responses for the R1 checkpoints are likewise unpublished, while the open-source baselines’ OlympiadBench responses are published and re-judged above. Regenerating the R1 seed-42 OlympiadBench responses greedily from the released checkpoint and scoring them with the local judge gives 52.2% (361/692), against the original run’s Sonnet-judged 45.4%; the implied margin sits inside OlympiadBench’s observed leniency band, and the local-judge model ordering matches Sonnet’s. The local judge is more lenient than Sonnet in every column; the largest gap (+30.3 pp) is on the dense-reward ablation’s PhysReason responses, consistent with the format-proxy channel of §4: dense-shaped rollouts produce well-formatted near-miss answers that a lenient judge accepts. Per-record local verdicts are released alongside the Sonnet verdicts (localjudge-qwen3-32b/ in the eval-outputs dataset). One provenance caveat: the InternVL3-8B OlympiadBench row re-judges the earlier same-budget generation run (Sonnet-judged 13.4 on that response set), not the later run behind Table 2’s 10.4 cell, so its delta is internal to that response set.

Additional open-source baseline: LLaVA-OneVision-7B. To extend the open-source block beyond the InternVL3/Qwen3-VL pair, we evaluated LLaVA-OneVision-7B [Li et al., 2024] (SigLIP vision encoder, Qwen2 language backbone, LLaVA training lineage) across the full Table 2 row at the 16,384-token budget. The PhyX cells are standard multimodal MCQ exact-match. The open-ended cells in Table 2 report local-judge liberal accuracy; the corresponding strict scores are PhysReason 0.8, PUB-OE 4.3, OlympiadBench-Physics 0.0, and PHYSOLYM-A 0.0. PhysReason is evaluated with images; the remaining open-ended columns are text-only, because a vLLM multi-modal kernel defect blocked image input for this architecture on those sets (disclosed here so the cells are not read as full-protocol results). On PHYSOLYM-A the protocol is the identical text-only one used for all models. It emits a boxed final answer on 137/500 problems and scores 0.0% strict; under the local open-weight judge it scores 7.2% liberal, and since that judge is +14.6 pp more lenient than Sonnet on this benchmark, its Sonnet-judged score would fall at or below InternVL3-8B’s 4.0%. This is consistent with the main-text finding that novel-source open-ended olympiad evaluation is far below the saturated-MCQ regime for open-source models at this scale. Raw responses and verdicts are released with the paper (llava-onevision-7b/ in the eval-outputs dataset).

Per-chunk verdict breakdown for PHYSOLYM-A (Sonnet 4.5). The 500 problems are partitioned into 5 chunks of 100 (random seed 42, source-stratified). Per-chunk liberal (judge-YES) counts: 38, 20, 35, 30, 40 over judgeable-only denominators 98, 100, 99, 98, 96 (491 of 500; the 9 dropped records, 1.8%, are those where the judge call itself failed, distinct from the 13.9% gold-side

Table 4: The local open-weight judge (Qwen3-32B, identical templates) is more lenient than Sonnet in every re-derivable cell, but by a model-dependent margin (+1.5 to +37.1 pp): absolute numbers are conservative under Sonnet, while cross-model gaps can narrow or reverse under a different judge, so model comparisons should be read under one fixed judge. Problem-level agreement vs. the pinned Sonnet verdicts; the local judge also scores the 13.9% of PHYSOLYM-A golds that Sonnet’s protocol marks unjudgeable, which contributes to the larger PHYSOLYM-A deltas.

Column	Checkpoint	n	Agree	κ	Local / Sonnet	Δ (pp)
PHYSOLYM-A	binary s42	500	80.2%	0.56	40.2 / 25.6	+14.6
PhysReason	binary s42	1258	76.6%	0.54	51.9 / 32.2	+19.7
PhysReason	binary s17	1200	85.4%	0.71	53.3 / 43.1	+10.3
PhysReason	binary s23	1200	85.8%	0.72	53.8 / 43.4	+10.3
PhysReason	dense s42	1200	67.9%	0.38	53.5 / 23.3	+30.3
PUB-OE	binary s42	629	81.4%	0.63	52.5 / 37.0	+15.4
PUB-OE	binary s17	629	78.9%	0.59	55.0 / 36.1	+18.9
PUB-OE	binary s23	629	81.4%	0.62	46.1 / 30.4	+15.7
PUB-OE	dense s42	629	79.3%	0.60	56.1 / 37.7	+18.4
PhysReason	Qwen3-VL-8B base	1200	73.5%	0.46	49.1 / 23.9	+25.2
PUB-OE	Qwen3-VL-8B base	629	79.2%	0.59	52.3 / 35.3	+17.0
PHYSOLYM-A	Qwen3-VL-8B base	500	81.0%	0.38	27.0 / 8.0	+19.0
OlymBench	Qwen3-VL-8B base	692	91.3%	0.82	40.8 / 39.3	+1.5
PhysReason	Qwen3-VL-32B	1200	60.9%	0.30	62.2 / 25.1	+37.1
PUB-OE	Qwen3-VL-32B	629	76.0%	0.53	54.2 / 32.8	+21.5
PHYSOLYM-A	Qwen3-VL-32B	500	68.0%	0.31	45.2 / 13.2	+32.0
OlymBench	Qwen3-VL-32B	692	89.5%	0.79	57.5 / 53.9	+3.6
PhysReason	InternVL3-8B	1200	89.5%	0.65	22.9 / 13.3	+9.7
PUB-OE	InternVL3-8B	629	82.0%	0.59	37.7 / 23.5	+14.2
PHYSOLYM-A	InternVL3-8B	500	80.8%	0.24	23.2 / 4.0	+19.2
OlymBench	InternVL3-8B	692	89.7%	0.65	21.7 / 13.4	+8.2

unjudgeable rate of §5.1). Liberal accuracy over judged records is $163/491 = 33.2\%$. The chunk-1 outlier at 20% accuracy is concentrated in reference-pointer Zhou problems whose gold solutions cite external olympiad-handbook material rather than providing self-contained answers; these are the unjudgeable category we surface as a known noise floor.

Inter-judge agreement. The Sonnet judge is run with two seeds on the same 500 problems for PHYSOLYM-A. Cohen’s κ on the YES/NO label between the two passes is reported alongside the released dataset; preliminary inspection shows $\kappa \geq 0.8$ on the PHYSOLYM-A corpus.

Human-graded subset. A 100-problem random subsample of PHYSOLYM-A Sonnet predictions is graded by a single physics-trained annotator using the same YES/NO rubric. Per-record human-vs-LLM agreement and Cohen’s κ are released as a JSON alongside the dataset. We report this κ as a calibration check on the LLM judge, not as a calibrated human-baseline ground-truth (cf. Section 6).

Released artifacts. The verbatim judge prompts (problem-level + per-subpart), the YES/NO rubric, the per-chunk verdict breakdown, the 100-problem human-graded subset, and the per-record agreement matrices are released as `judge_prompts.txt`, `rubric.json`, `per_chunk_verdicts.json`, and `human_graded_subset.json`. The `pub_oe_gold_cache.json` (per-id clean per-subpart gold list) is released alongside `llm_judge_v3_pubeo.py` and is reproducible from the original PhysUniBench-OE source via the cache-builder script in the same directory.

Self-grading concern. Sonnet 4.5 is both the highest-scoring frontier baseline on PHYSOLYM-A and the judge used for liberal accuracy. Three checks bound any self-favoring bias: (i) the strict (numeric/symbolic match, judge-independent) Sonnet score is reported alongside liberal—the 4.7-pp Sonnet strict-vs-liberal gap (28.7% vs. 33.4%) bounds maximum leniency; (ii) on the failure-mode taxonomy (Appendix H.1) the dominant Sonnet error category is `wrong_subpart` (structural mismatch any judge marks wrong), not `valid_partial`; (iii) we report a cross-vendor judge agreement against GPT-4o on a 50-problem random subsample of PHYSOLYM-A (Qwen3-VL-8B-Thinking re-

sponses, seed 42). Sonnet 4.5 and GPT-4o under an identical prompt template show 88% **raw agreement** (44/50) and **Cohen’s** $\kappa=0.44$ (moderate, per Landis & Koch). The disagreement is asymmetric: GPT-4o flips 5 Sonnet-NO records to YES, while only 1 Sonnet-YES record is flipped by GPT-4o to NO (McNemar exact on 6 discordant pairs $\{5:\text{Sonnet-NO} \rightarrow \text{GPT-YES}, 1:\text{Sonnet-YES} \rightarrow \text{GPT-NO}\}$, two-sided $p=0.219$; the asymmetry is not significant at $n=6$, but the direction is preserved on a larger 200-problem ablation left to follow-up work). GPT-4o’s positive rate (16%, 8/50) is roughly *twice* Sonnet’s (8%, 4/50), which means the cross-vendor judge would assign Physics-R1 *higher* numbers than the Sonnet-judge headlines, not lower—the self-grading direction is the opposite of what the self-favoring concern would predict. The full per-pair verdicts are released as `cross_judge_50.jsonl` alongside the dataset.

E Per-source license and provenance log

Per-source provenance for each of the nine source families: full name, scrape URL, scrape date, original license string, redistribution tier, and outreach log.

UGPhysics. Source: Xu et al. [2025] (ICML 2025). 5,520 EN/ZH undergraduate physics problems. Scrape URL: <https://huggingface.co/datasets/UGPhysics/ugphysics-bench>. Scrape date: 2026-03-15. Original license: CC BY-NC-SA 4.0. Redistribution: CC BY-NC-SA 4.0 carried through. Outreach: not contacted (license is permissive for academic redistribution under same-license sharing).

OpenStax College + University Physics. Source: OpenStax (Rice University). 2,381 records harvested from end-of-chapter exercises. Scrape URL: <https://openstax.org/details/books/college-physics-2e> and [.../university-physics-volume-1](https://openstax.org/details/books/university-physics-volume-1). Scrape date: 2025-12-20. Original license: CC BY 4.0. Redistribution: CC BY 4.0 carried through; attribution to OpenStax preserved per record.

Physics Stack Exchange. Source: Physics Stack Exchange Q&A archive. 2,291 problem-and-accepted-answer records filtered for olympiad-style physics. Scrape URL: <https://physics.stackexchange.com/> via Stack Exchange data dump. Scrape date: 2026-01-08. Original license: CC BY-SA 4.0 (Stack Exchange contributor agreement). Redistribution: CC BY-SA 4.0 carried through.

MMMU + o1-CoT seed (RL-SFT seed). Source: 1,293-record pool of MMMU physics problems augmented with o1-style CoT solutions generated by Sonnet 4.5. MMMU base license: MIT [Yue et al., 2024]; generated CoT is our contribution. Scrape date: 2026-02-10. Redistribution: MIT (MMMU base) with generated CoT released under CC BY 4.0.

PhysReason. Source: Zhang et al. [2025] (ACL 2025). 1,200 step-graded physics reasoning problems. Scrape URL: <https://huggingface.co/datasets/PhysReason>. Scrape date: 2026-02-22. Original license: CC BY 4.0. Redistribution: CC BY 4.0 carried through.

Estonian Physics Olympiad collection. Source: Estonian Physics Olympiad (EFO), 2004–2018. 418 problems with organizer-issued 1–10 difficulty labels and a 201-problem bilingual EN+ET subset. Scrape URL: <https://fyysika.ee/> (rounds: lahtine, koolivoor, vabariikilik). Scrape date: 2026-03-30. Provenance: publicly archived problems and solutions on the official EFO portal, released by the Estonian Physics Olympiad committee for educational use under competition policy (consistent with international physics-olympiad practice for IPhO, NBPhO, EuPhO, APhO, US-APhO, INPhO). Redistribution: public-domain by competition policy, used for non-commercial research evaluation; downstream users redistributing for commercial purposes or in materially modified form should consult <https://fyysika.ee/> directly.

Kevin Zhou’s olympiad handouts. Source: Kevin Zhou’s olympiad training documents (US-APhO/IPhO/APhO/EuPhO content with Cambridge Tripos and graduate-qualifier extensions), <https://knzhou.github.io/>. 692 problems with native point values 1–5 and a 3.2% advanced [A] flag. Scrape date: 2026-02-01. Original license: CC BY-NC 4.0 (written confirmation from Kevin Zhou, archived in supplementary as `zhou_license_2026-05-02.eml`, SHA-

256 7f25c859f5ae1d790e45dbdf d23ab6be27aa1814a76de10f9bdbfffb67088aba4). Redistribution: the 692 redistributed problems plus their reference solutions ($\sim 1,600$ problem-solution items in total counting solutions as separate documents) are released under CC BY-NC 4.0 with attribution link to <https://knzhou.github.io/> preserved per record. Third-party content disclosure (per Zhou’s reply): some problems in the handouts are drawn from books or other olympiad archives, with the original source attributed inline by Zhou. We preserve all such per-problem internal attributions verbatim in the released records; downstream users should treat any in-record secondary-source attribution as binding under the original source’s terms, which may be more restrictive than CC BY-NC 4.0. Outreach log: initial inquiry 2026-04-10 (proposal: $\sim 1,600$ problems + solutions, CC BY-NC 4.0, attribution to source site); written agreement to the proposed terms 2026-05-03 with the third-party-content caveat noted by Zhou.

IPhO + NBPhO + EuPhO scrape. Sources: International Physics Olympiad (ipho-new.org), Nordic Baltic Physics Olympiad, European Physics Olympiad official archives. 258 problems (IPhO 66, NBPhO 165, EuPhO 27). Scrape date: 2026-03-05. Original license: public-domain by competition policy (problems published for educational use without restriction; we preserve attribution per record). Redistribution: public-domain carried through with per-record attribution.

APhO + USAPhO + INPhO scrape. Sources: Asian Physics Olympiad, USA Physics Olympiad (Physics Olympiad Foundation), Indian National Physics Olympiad. 241 problems recovered after a multi-pattern splitter fix that recognizes A1.-style markers (USAPhO 1997–2006) and 1. (a) solution markers (INPhO). The INPhO recovery alone added +33 records relative to a single-regex baseline. Scrape date: 2026-03-12. Original license: public-domain by competition policy. Redistribution: public-domain carried through with per-record attribution.

Table 5: Per-source license and provenance. Each released record carries its source license through; non-commercial sources (UGPhysics, Zhou) restrict downstream use to academic research, and the public-domain-by-competition-policy olympiad scrapes (EFO, IPhO, NBPhO, EuPhO, APhO, USAPhO, INPhO) preserve per-record source attribution.

Source family	Records	Original license
OpenStax College + University Physics	2,381	CC BY 4.0
Physics Stack Exchange	2,291	CC BY-SA 4.0
PhysReason	1,200	CC BY 4.0
MMMU + o1-CoT seed (RL-SFT seed)	1,293	MIT (MMMU); generated CoT
IPhO + NBPhO + EuPhO scrape	258	Public-domain (competition policy)
APhO + USAPhO + INPhO scrape	241	Public-domain (competition policy)
UGPhysics	5,520	CC BY-NC-SA 4.0
Estonian Physics Olympiad	418	Public-domain (competition policy)
Kevin Zhou’s olympiad handouts	692	CC BY-NC 4.0 (Zhou confirmed 2026-05-03)
Total pre-audit	14,294	

F Reproducibility checklist

Experiment	Wall-clock	Hardware	Cost	Notes
Sonnet baselines (PhyX / OlymBench / PHYSOLYM-A)	~ 5 h	API	$\sim \$80$	Two-judge
Open-source baseline sweep on PhyX-mini-MC 1000q	~ 90 min	1 $\times$ H200	$\sim \$5$	vLLM 0.11.0 bf16
Contamination audit, Stages 1–2	~ 3 min	MPS/CPU	free	5 k records pairwise; Stage-3 judge replayed offline from release
Threshold-sensitivity analysis	~ 3 min	MPS	free	sentence-transformers
Physics-R1 RL training to step 60+	~ 30 h	4 $\times$ H200	$\sim \$120$	verl 0.6.1, FSDP1
Reward-component drop-out (follow-up work)	~ 40 h	4 $\times$ H200	$\sim \$700$	Table 12
3-seed Physics-R1 sensitivity (seeds 17 + 23 retrainings)	~ 60 h	4 $\times$ H200	$\sim \$200$	added to seed-42 in Table 2
Total (this paper)			$\sim \$700$	seeds 42/17/23 + Sonnet baselines + Sonnet-judge runs
Follow-up budget (reward drop-out)			$\sim \$1,000$	Table 12

Compute budget per experiment.

Random seeds. The headline single-seed Physics-R1 binary checkpoint uses seed 42. The 3-seed mean reported in Table 2 aggregates seeds {42, 17, 23}, all on the audited PHYSR1CORP corpus under the binary correctness reward; checkpoint selection per seed uses MM-Eureka difficulty-curriculum saturation on the held-out PhyX-mini-MC (1,000-problem) early-stop signal. The data-build pipeline (PHYSOLYM-A sampling, train/val splits, audit pass) uses `numpy.random.default_rng(42)`.

Framework reproducibility note. Training uses `verl` 0.6.1 [Sheng et al., 2024] with FSDP1 sharding for Qwen3-VL: FSDP2 fails on the multimodal projector path in this version, so FSDP1 is required; the upstream issue link is tracked in the released README.

Version pins. `transformers == 4.57.0` (re-evaluation with 4.57.6 drifts results by ~ 1.5 points; we pin to 4.57.0 for reproducibility), `vllm == 0.11.0`, `verl == 0.6.1`, `sympy == 1.13.3`, `sentence-transformers == 5.4.1`, `torch == 2.8.0+cu128`. Embedding model: `mxbai-embed-large-v1` from MixedBread AI.

Code and data hosting. Code: <https://github.com/shanyang-me/physics-r1-neurips2026>. Datasets: <https://huggingface.co/datasets/shanyangmie/physolym-a> (eval split) and <https://huggingface.co/datasets/shanyangmie/physics-r1-corpus> (audit-clean training pool). Raw evaluation outputs and per-record judge verdicts: <https://huggingface.co/datasets/shanyangmie/physics-r1-eval-outputs>. Croissant 1.0 metadata is auto-generated by HuggingFace at `/api/datasets/<repo>/croissant` for each dataset, then augmented with Responsible AI (RAI) metadata fields via the NeurIPS-recommended RAI editor (<https://huggingface.co/spaces/JoaquinVanschoren/croissant-rai-checker>) and validated with the Croissant validator (<https://huggingface.co/spaces/JoaquinVanschoren/croissant-checker>); the RAI-augmented files (`croissant_rai_<artifact>.json`) ship in the supplementary archive. All four artifacts (PHYSCORP-PRE-AUDIT, PHYSCORP-A, PHYSR1CORP, PHYSOLYM-A) are released alongside this paper with all nine source licenses confirmed in writing (Appendix E).

Quick-start audit.

```
python audit/audit_two_stage.py \
  --train_jsonl your_pool.jsonl --eval_jsonl data/physolym_a.jsonl \
  --jaccard_thr 0.4 --cosine_thr 0.85 --emit report.json
```

Writes per-record audit + aggregate 3×3 threshold-sensitivity table (Table 3); ~ 3 min on MPS or a CUDA GPU for a 5,000-record pool against 4 held-out splits, with shingle/embedding caches in `audit_cache/`. Stage-3 is applied by replaying the released per-pair `judge_label` arrays (no API access needed); re-running the judge live uses the released prompt template and pinned model id (Appendix A).

Supplementary materials index. (i) Croissant 1.0 + RAI JSON-LD per artifact (`croissant_rai_<artifact>.json`); (ii) the four released datasets (`physcorp_a.jsonl`, `physr1corp.jsonl`, `physolym_a.jsonl`, plus PHYSCORP-PRE-AUDIT); (iii) audit pipeline (`audit_two_stage.py` + `threshold_sensitivity_scores.npz` with per-pair `judge_label` arrays and the Stage-3 prompt template); (iv) reward implementation (`reward_physics.py`); (v) LLM-judge artifacts (`judge_prompt.txt`, `rubric.json`, `per_chunk_verdicts.json`, `human_graded_subset.json`); (vi) archived license confirmation (`zhou_license_2026-05-02.eml`, the Kevin Zhou CC BY-NC 4.0 grant); (vii) training config (`configs/physics-r1.yaml`); (viii) checkpoints at steps {20, 40, 60, 80}.

Reproducibility checklist. Code released with `requirements.txt` and deterministic build script; data released with per-source license provenance (Table 5) and Croissant 1.0+RAI metadata; datasheet in Appendix G; compute, seeds, and hyperparameters above + Table 11. Hosting: HuggingFace (≥ 5 yr) + GitHub + Zenodo. Maintenance: quarterly contamination audit. Follow-up commitments: reward-component drop-out ablation (Table 12), embedder-sensitivity audit against `voyage-3` and `text-embedding-3-large`, paraphrase/translation-aware audit pass.

Recommended baseline configuration. Algorithm 3 captures the joint setting; each individual choice is small in magnitude. Binary correctness is the recommended default; dense (Algorithm 2) is reported as an ablation.

Algorithm 3 Recommended baseline configuration for physics-VL RL post-training.

Require: Base thinking-mode VLM π_{base} , audited training pool T' (Algorithm 1), held-out MCQ early-stop signal H_{MC} , dense reward r (Algorithm 2)

- 1: Init: $\pi_{\theta} \leftarrow \pi_{\text{base}}$ ▷ cold-start from base, no SFT pass (MM-Eureka thesis)
- 2: Optimizer: GSPO+DAPO (sequence-level importance, decoupled clip), unmodified
- 3: KL anchor: add $\beta_{\text{KL}} D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{base}})$ with $\beta_{\text{KL}} = 10^{-3}$ ▷ bounds drift
- 4: Entropy bonus: add $\beta_H \mathcal{H}(\pi_{\theta})$ with $\beta_H = 10^{-3}$ ▷ prevents entropy collapse
- 5: Difficulty curriculum: drop train items where the base model gets 0/ N or N / N rollouts ▷ MM-Eureka-style; preserves learning signal
- 6: LR schedule: 1×10^{-6} initial, step-cosine decay; halve LR after the first plateau on H_{MC} ▷ counters drift and length collapse
- 7: Response budget: fixed long-CoT budget (12,288 tokens for thinking-mode), *not adaptive* ▷ stable per-step cost
- 8: Early stopping: stop at the saturation peak on held-out MCQ H_{MC} , *not* on the train-distribution validation set ▷ addresses train-up/eval-down divergence
- 9: Reward: *recommended* binary correctness reward $r_{\text{ans}} \in \{0, +1\}$ (simpler, fully reproducible on v11m == 0.11.0 multi-image eval); dense five-component physics-native reward r (Algorithm 2) is reported as an ablation
- 10: Audit gate: train pool must be Algorithm 1-audited against held-out splits *and* external benchmarks before training begins
- 11: Reproducibility pins: transformers == 4.57.0, v11m == 0.11.0, verl == 0.6.1, FSDP1 sharding for Qwen3-VL (Appendix F)

G Datasheet for Physics-R1

We follow Gebru et al. [2021] with seven sections: Motivation, Composition, Collection process, Preprocessing/cleaning/labeling, Uses, Distribution, and Maintenance. The full per-source provenance log is in Appendix E; the audit pipeline in Appendix A; the LLM-judge protocol in Appendix D.

G.1 Motivation

For what purpose was the dataset created? To support contamination-audited evaluation and post-training of multimodal vision-language models on visual physics reasoning, with three specific gaps the field had not closed: (i) no public physics-VL training pool was audited under a three-stage (n-gram, embedding, LLM-judge) protocol that catches paraphrase-class duplicates and recovers threshold-edge topic-similarity false positives; (ii) no public physics-olympiad eval was both novel-source and contamination-clean against the major training-side aggregations (PhyX, MMMU-Pro Physics, OlympiadBench-Physics, UGPhysics); (iii) no public physics-VL benchmark exposed the format-and-novelty saturation gradient at frontier-model scale. Who created the dataset and on whose behalf? Shan Yang. Who funded the creation? Self-funded; no third-party sponsor. Other comments. The corpus aggregates and audits material that already existed in scattered formats; the Estonian Physics Olympiad, Kevin Zhou’s olympiad handouts, and six international olympiads are first-time ML-format releases.

G.2 Composition

Instances: one physics problem per record (statement, optional images PNG/JPEG, gold MCQ-letter / numeric / symbolic / multi-part answer, optional reference solution, eight-field annotation; schema in §B). *Counts:* PHYSCORP-A 6,432 audited; PHYSR1CORP 2,268 closed-form RL pool; PHYSOLYM-A 500 held-out (stratified sample, seed 42, source-family-stratified); PHYSCORP-PRE-AUDIT 14,294 raw. *Labels:* gold answer + schema labels; ~3,900 records carry full

Sonnet-4.5 annotation, rest from source-native labels; `native_difficulty` present where organizers publish (Estonian 27%, Zhou 38%). *Splits*: see §B. *Noise*: LLM-judge unjudgeable rate 13.9% on PHYSOLYM-A; Stage-1 audit misses paraphrase/translation, Stage-2 misses numerical substitution—both reported as floors. *Self-contained*: yes for problems and solutions; some Zhou records carry inline secondary-source attribution (Appendix E). *No PII or sensitive content*.

G.3 Collection process

Acquisition: 5 repackaged benchmark releases (UGPhysics, OpenStax, Physics Stack Exchange, MMMU+o1-CoT, PhysReason) + 4 first-ML scrapes by authors (Estonian PhO, Kevin Zhou’s handouts, 7 international olympiads); per-source URLs and dates in Appendix E. *Sampling*: per-source complete enumeration over public archive date ranges. *Authorship*: the author (Shan Yang) handled scraping/parsing/audit; Sonnet 4.5 batch produced annotation labels. *Time frame*: scrapes 2025-12 to 2026-04, audit/curation 2026-02 to 2026-04. *Ethics*: no human subjects, no PII, no IRB. *Consent*: Kevin Zhou confirmed CC BY-NC 4.0 redistribution of his olympiad handouts in writing 2026-05-03; the Estonian Physics Olympiad and other olympiad sources (IPhO, NBPhO, EuPhO, APhO, USAPhO, INPhO) are released under public-domain competition policy with per-record source attribution; repackaged sources are redistributed under their original CC/MIT licenses with attribution preserved.

G.4 Preprocessing, cleaning, labeling

Pre-tokenization normalization for Stage-1 audit (Appendix A); three-stage audit (Stage-1 $J \geq 0.4$, Stage-2 $\cos \geq 0.85$, Stage-3 Haiku-4.5 LLM-judge close-duplicate vs. same-topic-neighbor classification) pairwise across PhyX, MMMU-Pro Physics, OlympiadBench-Physics, UGPhysics-Train, PHYSOLYM-A, with Stage-3 close-duplicate records removed; eight-field schema annotation by Sonnet 4.5 batch (3,900 records) + source-native labels. Raw pre-audit pool (PHYSCORP-PRE-AUDIT, 14,294 records) released alongside so users can re-run audits. Audit pipeline released as `audit_two_stage.py` plus the Stage-3 judge artifacts: saved `best_jaccard/best_cosine/judge_label` arrays and the judge prompt template.

G.5 Uses

Used for: Physics-R1 RL recipe (§4) trains on the audited pool and evaluates on PHYSOLYM-A + auxiliary splits. Supplementary archive ships paper, datasets, Croissant+RAI metadata, .eml license confirmations, checkpoints. *Other use cases*: visual physics reasoning eval, contamination-audit methodology, cross-lingual studies (EN/ET subset), native-difficulty calibration, VLM RL post-training. *Caveats*: cross-lingual finding is Sonnet-specific; dense reward is an ablation not a tuned standard. *Out of scope*: general physics ability beyond visual reasoning, human olympiad grading substitution, experimental physics, research capability; held-out splits must not enter pretraining/fine-tuning.

G.6 Distribution

Distribution: HuggingFace (≥ 5 yr) + GitHub + Zenodo DOI. URLs: `huggingface.co/datasets/shanyangmie/physolym-a` and `huggingface.co/datasets/shanyangmie/physics-r1-corpus`; code at `github.com/shanyang-me/physics-r1-neurips2026`. All four artifacts released alongside this paper with per-source licenses documented in Appendix E; the Kevin Zhou CC BY-NC 4.0 grant is preserved as a written .eml in the supplementary archive, and the remaining olympiad sources are released under public-domain competition policy. *Licenses*: CC BY 4.0, CC BY-SA 4.0, public-domain by competition policy (EFO, IPhO, NBPhO, EuPhO, APhO, USAPhO, INPhO), MIT, CC BY-NC 4.0 (Zhou), CC BY-NC-SA 4.0 (UGPhysics)—each record carries its source license; Table 5. CC BY-NC sources are non-commercial; Zhou records honor inline secondary-source attribution. No export controls.

G.7 Maintenance

Maintained by the author (Shan Yang, `alexxyangshan@gmail.com`); contact via email or GitHub Issues at `github.com/shanyang-me/physics-r1-neurips2026`. Versioned CHANGELOG

(v1.0.0 initial release, v1.1.x additive, v2.0.0 schema-breaking); per-record diffs per release. Quarterly contamination audit against new physics-VL benchmarks; $\geq 1\%$ leakage triggers documented-diff removal. Planned follow-ups: paraphrase/translation-aware audit, embedder-sensitivity ablation against voyage-3 / text-embedding-3-large. Hosting ≥ 5 yr on HF + Zenodo DOI per release; all versions tagged and accessible. Contributions via GitHub Issues / PR + per-record erratum.json, reviewed within 60 days.

Machine-readable metadata: **Croissant + RAI JSON-LD.** The release ships a Croissant 1.0 [Akhtar et al., 2024] JSON-LD descriptor (croissant.json) declaring distribution objects for PHYSRICORP, PHYSOLYM-A, and the audit-pipeline source archive, plus a eight-field problem_record schema and the full RAI extension (rai:dataCollection, rai:dataAnnotationProtocol, rai:dataPreprocessingProtocol, rai:personalSensitiveInformation, rai:dataLimitations, rai:dataReleaseMaintenancePlan, rai:dataUseCases, rai:dataBiases, rai:dataSocialImpact) covering the audit methodology, the Kevin Zhou CC BY-NC 4.0 written grant (Appendix E), the public-domain-by-competition-policy basis for the olympiad scrapes, three documented distributional biases (EN/ET asymmetry, per-physics-category variance, difficulty-stratified decay), and the Stage-1/Stage-2 thresholds. Passes mlcroissant validate and the MLCommons RAI checker.

H Extended discussion: failure modes, ethics, methodological notes, and future work

This appendix collects extended-discussion content that was trimmed from the main body to fit the page limit. Each subsection below corresponds to a one-line pointer in the analysis section (§6).

H.1 Failure-mode taxonomy (extended)

A manual taxonomy of 100 randomly-sampled wrong or partial Sonnet predictions on OlympiadBench-Physics was completed post-evaluation. Methodology. The 100 cases were drawn (random seed 42) from the 354 predictions flagged wrong or partial by the scoring pass used at annotation time; each case was categorized by a single physics-trained annotator (graduate-level physics background, ~ 3 hours of total annotation time, ~ 1.8 min/case median) using a nine-category mutually-exclusive rubric (wrong_subpart, missing_physics, calc_error, different_question, valid_partial, symbolic_vs_numeric, magnitude_error, sign_error, diagram_misread). The annotator saw the problem statement, the gold solution, the gold final answer, and the Sonnet response; they did not see Sonnet’s confidence or the verdict from the LLM judge. We report this single-annotator taxonomy as a methodological diagnostic, *not* as a calibrated human grade; a second-annotator pass on a 50-problem random subsample with inter-annotator κ per category is left to follow-up work (distinct from the audit-flag inter-annotator κ pre-registered in Appendix A, which targets contamination-flag agreement, not failure-category agreement).

Table 6: Failure-mode taxonomy of 100 randomly-sampled Sonnet wrong/partial predictions on OlympiadBench-Physics. Categories are mutually exclusive; each prediction received one label.

Category	n	%	Description
wrong_subpart	30	30%	Answered a neighboring sub-question instead of the one asked
missing_physics	22	22%	Wrong physical model, law, or geometry; crucial effect absent
calc_error	16	16%	Correct approach; small numerical slip (factor 2–4 off)
different_question	10	10%	Reasoning chain for an unrelated problem in the same prompt
valid_partial	8	8%	Right approach with arithmetic slip; partial-credit territory
symbolic_vs_numeric	7	7%	Formula returned where a number was required, or vice versa
magnitude_error	5	5%	Correct expression, wrong order of magnitude ($\geq 10\times$)
sign_error	0	0%	None observed
diagram_misread	0	0%	None observed (text-only evaluation; figures unavailable)
Total	100	100%	

Reading the taxonomy. `wrong_subpart` dominates (30%): multi-part olympiad prompts containing (a)/(b)/(c) sub-questions where the model fluently answers a neighboring sub-question instead of the asked one. `missing_physics` (22%) and `different_question` (10%) together account for nearly a third of failures and represent a real reasoning floor unlikely to be format-induced. `calc_error` (16%) and `valid_partial` (8%) are correct-approach/failed-execution, matching the 4.7-pp strict-vs-liberal gap (§5.2). `sign_error` and `diagram_misread` are zero (the latter because OlympiadBench-Physics is text-only in the public release).

Table 7: Per-physics-category accuracy on OlympiadBench-Physics (Sonnet 4.5, strict). The +34.5-pp EM (38.4%) vs. astrophysics (72.9%) gap on identical weights motivates per-category reporting.

Category	n	Acc.	Category	n	Acc.	Category	n	Acc.
Electromagnetism	237	38.4%	Relativity	89	51.7%	Other	10	60.0%
Quantum	128	41.4%	Classical mech.	19	57.9%	Astrophysics	70	72.9%
Waves	39	46.2%	Thermodynamics	87	59.8%	All	692	—
Fluid mechanics	13	46.2%						

Per-physics-category breakdown.

H.2 Ethical considerations and intended use

Intended and out-of-scope use. The released artifacts support research on visual physics reasoning, contamination-audit methodology, cross-lingual LLM evaluation, and RL post-training of VLMs. PHYSOLYM-A is held-out: please treat it as test-only, not for pretraining or fine-tuning. Out of scope: general physics ability beyond visual reasoning, human olympiad grading substitution, experimental-physics evaluation, paper-writing, derivation novelty, or open-ended hypothesis generation.

Source provenance, consent, and misuse risk. Kevin Zhou confirmed CC BY-NC 4.0 redistribution of his olympiad handouts in writing on 2026-05-03 (~1,600 problem-solution items, Appendix E); the Estonian Physics Olympiad collection and the six international olympiad scrapes (IPhO, NBPhO, EuPhO, APhO, USAPhO, INPhO) are redistributed under public-domain competition policy with per-record source attribution. No PII; problems are olympiad/textbook content. Public-domain olympiad scrapes are released by competition policy. The dense reward and Physics-R1 recipe ship for reproducibility, not as a tuned gold standard; the audit pipeline is a measurement tool, not a certification authority, and we disclose threshold sensitivity (Table 3) so users do not over-claim “contamination-free.”

Caveats and compute. The 17-pp EN/ET cross-lingual delta is Sonnet-4.5-specific on $n=59$ paired items (McNemar $p=0.021$; bootstrap 95% CI [+5.1, +28.9] pp); direction may flip on low-resource-language-weak open-source models—treat as model-specific. Total compute ~360 H200-GPU-hours across the 3 seeds in Table 2 (seed-42 at 30 h $\times$ 4 H200; seed-17 and seed-23 retrains $\sim$ 60 h $\times$ 4 H200 combined; Appendix F), plus frontier-API inference for baselines and Sonnet-judge runs. Per-experiment carbon estimates are omitted because cloud-provider electricity mix is not consistently disclosed.

H.3 Construction-process disclosures

(i) The initial PHYSOLYM-A description claimed 100% novel-source; the Stage-1 audit surfaced one EuPhO 2020 overlap with OlympiadBench-Physics ($J=0.91$), so the honest claim is 99.8% (499/500)—disclosed, not dropped. (ii) Our initial header described PhyX-mini-MC as 500 problems; the canonical MC subset [Shen et al., 2025] is 1,000.

H.4 Future work and named follow-ups

Ten follow-ups consolidated. *Eval refinements:* (i) post-hoc MCQ-ification of PHYSOLYM-A to isolate the format axis (§5.2); (ii) paraphrase- and translation-aware audit pass. *Cross-corpus:* (iii) audit OlympiadBench-Physics against our pool, then add the Physics-R1 row; (iv) frontier

Table 8: **Physics-R1 vs. rule-based RL recipes for thinking-mode VLMs.** Init: base or SFT cold-start. Reward signals: *Ans* (answer), *Fmt* (format), *Dim* (units), *Sym* (symbolic), *Cons* (conservation). Audit and Filter as defined in §3.1. ✓ present; — absent; ◦ partial.

Recipe	Init	Ans	Fmt	Dim	Sym	Cons	Audit	Filter
DPO [Rafailov et al., 2023]	SFT	—	—	—	—	—	—	—
GRPO [Shao et al., 2024]	SFT	✓	—	—	—	—	—	—
GSPO+DAPO [Zheng et al., 2025, Yu et al., 2025]	SFT	✓	◦	—	—	—	—	—
MM-Eureka [Meng et al., 2025]	base	✓	✓	—	—	—	—	—
Physics-R1	base	✓	✓	✓	✓	✓	✓	✓

cross-evaluation against PHYBench, PhysUniBench, HLE-physics, PHYSOLYM-A. *Recipe and scale:* (v) transfer to InternVL3-8B and LLaVA-OneVision-7B; (vi) 32B + full-RL comparator; (vii) SFT-only data-scaling curve at 500/1,293/5,000/9,575 audited prompts. *Cross-lingual:* (viii) confirm/refute EN/ET sign-flip on low-resource open-source models on the same 59-pair Estonian Physics Olympiad subset; *pre-registered*—if 8B-class open-source models with documented Estonian-weak training (e.g. Qwen2.5-VL-7B, LLaVA-OV-7B) show ET<EN by ≥ 5 pp under paired sign test, F2’s Sonnet-4.5 ET>EN direction is model-specific (confirmed); a ≥ 5 pp result in the same direction (ET>EN) on those same models would replicate F2 across model families; results within ± 5 pp are inconclusive at $n=59$. *Methodology:* (ix) 50-problem inter-annotator κ on the failure-mode taxonomy; (x) embedder-sensitivity ablation against voyage-3 and text-embedding-3-large.

H.5 Versioning and maintenance commitment

Versioned releases with semantic-version tags (v1.0.0 initial, v1.1.x additive, v2.0.0 schema-breaking). Each release ships Croissant 1.0 + RAI JSON-LD, per-source license matrix (Table 5), threshold-sensitivity grid (Table 3) recomputed against newly-released physics-VL benchmarks, and the audit-pipeline source archive with saved best-overlap scores. Quarterly contamination audit against new benchmarks; benchmarks introducing $\geq 1\%$ leakage to any held-out split trigger a documented-diff removal in the next minor release. Hosting: HuggingFace (≥ 5 yr), GitHub, Zenodo DOI per release.

H.6 Method comparison and benchmark comparison tables (extended)

Table 8 situates the recipe configuration against related rule-based RL recipes; its dimensional/symbolic/conservation reward columns describe the dense *ablation* only, not the recommended binary default.

H.7 Corpus composition, hyperparameters, and reward ablation (extended)

Per-seed values behind the Table 2 3-seed row. Seeds 42/17/23: PhysReason 32.2/43.1/43.4; PUB-OE 37.0/36.1/30.4; OlympiadBench-Physics 45.4/45.3/48.0; PHYSOLYM-A 25.6/25.0/28.2; PhyX-1k 78.0/77.4/77.9; PhyX-3k 76.9/77.2/76.6. Seed 42 is also the dense-ablation seed. The Sonnet PhysReason baseline cell in Table 2 is regenerated at `max_tokens = 16384` with 1,192/1,200 clean records.

Table 9: **Released artifacts vs related benchmarks across nine axes.** *Audit:* 3-stage (n-gram+embedding+LLM judge) / 2-stage / 1-stage / orig. (constructed-novel) / none. *T/T leak:* train→test joint-stage ($J \geq 0.4 \vee \cos \geq 0.85$) audit against six public physics evals; ✓ all 6 = clean. *Diff:* organizer difficulty. *X-L:* paired cross-lingual. *Use:* E/T = eval/train. *RL-ready:* closed-form gold + audit-clean + RL recipe. “.” = eval-only; “n/r” = train pool, no cross-corpus audit. Only this work reports train/test contamination: after re-audit cleanup, PHYSCORP-A (6,432) and PHYSR1CORP (2,268) are clean against **all six** evals (Table 1).

Benchmark	Size	Format	MM	Audit	T/T leak	Diff	X-L	Use	RL-ready
<i>Physics-domain benchmarks</i>									
PHYBench [Qiu et al., 2025]	500	open+EED	—	orig.	.	—	—	E	—
PhysUniBench [Wang et al., 2025b]	3,304	open MM	✓	1-stage	.	✓	—	E	—
UGPhysics [Xu et al., 2025]	5,520	open text	—	1-stage	n/r	—	EN/ZH	T	—
PhysReason [Zhang et al., 2025]	1,200	step open MM	✓	none	.	—	—	E	—
OlympiadBench [He et al., 2024]	8,476	open MM	✓	none	.	—	EN/ZH	E	—
<i>Olympiad / formal / contamination-by-design</i>									
PutnamBench [Tsoukalas et al., 2024]	1,692	Lean/Isab.	—	orig.	.	✓	—	E	—
OIBench [Zhu et al., 2025b]	250	open code	—	2-stage	.	✓	EN/ZH	E	—
FrontierMath [Glazer et al., 2024]	290	open math	—	orig.	.	✓	—	E	—
HLE [Phan et al., 2025]	2,500	expert exam	✓	orig.	.	—	—	E	—
<i>Multimodal / multi-domain</i>									
MMLU-Pro [Wang et al., 2024]	12,032	10-MCQ	—	none	.	—	—	E	—
MMMU-Pro Phys [Yue et al., 2025]	60	10-MCQ MM	✓	none	.	—	—	E	—
SciInstruct [Zhang et al., 2024]	254,K	SFT instr.	—	1-stage	n/r	—	—	T	—
<i>This work (train→test cross-corpus audit reported; Table 1)</i>									
PHYSCORP-A (ours)	6,432	open+MCQ MM	✓	3-stage	✓ all 6	✓	✓	T	✓
PHYSR1CORP (ours)	2,268	MCQ + num MM	✓	3-stage	✓ all 6	✓	✓	T	✓
PHYSOlym-A (ours)	500	open MM novel	✓	3-stage	· (eval; clean)	✓	EN/ET	E	—

✓ = present; — = absent or not reported. “3-stage” audit = pairwise 5-gram-Jaccard, embedding-cosine, and LLM-judge precision filtering against external corpora and held-out splits.

Table 10: **PHYSCORP-PRE-AUDIT composition by source family** (14,294 records total before the three-stage audit). The audited release PHYSCORP-A (6,432 records) is the subset surviving Algorithm 1 after a re-audit pass against PhysReason-full and PhysUniBench-en dropped an additional 804 records from a 7,236-record candidate (the construction audit excluded those two corpora). The 804-record re-audit drop concentrates on PhysReason-cousin records (540) and PhysUniBench-en-cousin records (186), drawn predominantly from the *repackaged-benchmark* source families (UGPhysics / OpenStax / Physics SE / MMMU+o1-CoT seed / PhysReason); the four *first-ML-format* source families (Estonian PhO, Zhou’s handouts, the 7-international-olympiad scrapes) are minimally affected and retain their full 1,609-record contribution to PHYSCORP-A. All 9 source families remain represented in the released pool. Per-source licenses are carried through to each released record; full provenance and outreach log in Appendix E.

Source family	Count	License	Answer type
UGPhysics	5,520	CC BY-NC-SA	Open-ended numeric
OpenStax Physics	2,381	CC BY 4.0	Numeric
Physics Stack Exchange	2,291	CC BY-SA 4.0	Equations
RL-SFT seed (MMMU + o1 CoT)	1,293	MIT (MMMU); generated CoT	Varied
PhysReason	1,200	CC BY 4.0	CoT + open-ended
Zhou Olympiad Handouts	692	CC BY-NC 4.0 [†]	Mixed
Estonian Physics Olympiad	418	Public-domain (competition policy)	Open-ended
IPhO + NBPhO + EuPhO scrape	258	Public domain	Open-ended
APhO + USAPhO + INPhO scrape	241	Public domain	Open-ended
Total novel	1,609	—	—
Total corpus	14,294	—	—

[†] The handout author (K. Zhou) granted explicit redistribution permission with attribution (written confirmation, May 2026); we redistribute under CC BY-NC 4.0.

Table 11: **Physics-R1 training configuration.** GSPO+DAPO via ver1 0.6.1 with vLLM 0.11.0 (TP=4) for rollouts. FSDP1 is required for Qwen3-VL under ver1 0.6.1 (FSDP2 fails on the multimodal projector path; reproducibility note in Appendix F). Cells marked (*recipe*) differ from a default unconstrained GSPO+DAPO configuration; together with the binary correctness reward (Section 4; dense reported as an ablation), the audited training pool, and held-out early stopping, they constitute the Physics-R1 reference recipe.

Parameter	Value	Role in the recipe
Base / init (<i>recipe</i>)	Qwen3-VL-8B-Thinking BASE (<i>no SFT</i>)	Cold-start RL, mirroring MM-Eureka thesis
Algorithm	GSPO + DAPO	Stable group-policy backbone
Importance sampling (<i>recipe</i>)	truncated, sequence-level	Rollout-vs-train policy correction
Learning rate	1×10^{-6}	Standard for 8B-class RL
LR decay (<i>recipe</i>)	step-cosine; $0.5\times$ after step 60	Counters drift and length collapse
Batch size	96	Effective batch via gradient accumulation
Rollouts per prompt	16	DAPO group size
Max response length	12,288 tokens	Long-CoT thinking budget
KL anchor (<i>recipe</i>)	1×10^{-3} to base	Anchors policy to base; bounds drift
Entropy bonus (<i>recipe</i>)	1×10^{-3}	Prevents entropy collapse
Clip range (decoupled)	0.2 / 0.28	DAPO decoupled clip
Difficulty curriculum (<i>recipe</i>)	drop 0/16 and 16/16 rollout prompts	MM-Eureka-style; preserves learning signal
Reward (<i>recipe</i>)	binary correctness (§4)	Recommended default; dense (Algo. 2) is the shape ablation
Train pool (<i>recipe</i>)	2,268 PHYSR1CORP prompts	Closed-form carve-out of PHYSCORP-A (§3)
Early stop (<i>recipe</i>)	held-out PhyX-mini-MC	Catches the saturation peak
Sharding (<i>recipe</i>)	FSDP1 (not FSDP2)	Required for Qwen3-VL under ver1 0.6.1
Hardware	4×H200 (single node)	—
Step time	~40 minutes/step	Rollout-bound (gen ~65%, update ~23%)
Wall-clock to saturation peak	~30–45 hours (steps 60–80)	Training cost ~\$120–\$180
transformers	4.57.0 (pinned)	4.57.6 yields different forward path
vllm	0.11.0	TP=4 rollout backend
ver1	0.6.1	Training framework
Random seed (headline)	42	3-seed sweep {17, 23, 42} reported as the headline binary row in Table 2

Table 12: **Physics-R1 reward-shape ablation on PhyX-mini-MC.** Each row toggles one of the five reward components on or off, training Qwen3-VL-8B-Thinking-base under the same GSPO+DAPO+KL-anchor recipe (Equation 1) for the same step budget. Ans = answer-correctness binary (+1, $\equiv r_{\text{bin}}$ of §4); Fmt = `\boxed{}` format (+0.1); Dim = dimensional consistency from regex-detected units + sympy unit-system (+0.15); Sym = symbolic equation verification of intermediate `\frac` expressions via sympy (+0.20); Cons = conservation-law penalty (energy/momentum) when applicable (−0.25). Composed reward is clipped to $[-1, 1]$. Init in every cell is the Qwen3-VL-8B-Thinking BASE checkpoint (*no SFT*; cold-start, mirroring the MM-Eureka thesis). The Ans-only row is the recommended Physics-R1 recipe (binary correctness reward, §4); the all-on row is the dense ablation of §4; the intermediate rows isolate the marginal effect of each physics-native shaping component. Drop-out cells (—) are intermediate-component runs left to follow-up work (compute estimate in Appendix F).

Configuration	Ans	Fmt	Dim	Sym	Cons	PhyX-mini-MC
Base (no RL)	—	—	—	—	—	73.7%
Physics-R1 (binary, recommended) $\equiv$ Ans-only	✓	—	—	—	—	78.0
+ Format	✓	✓	—	—	—	—
+ Dim	✓	✓	✓	—	—	—
+ Sym	✓	✓	✓	✓	—	—
Physics-R1 (dense, ablation, all-on)	✓	✓	✓	✓	✓	78.3
<i>Single-component drop-outs (dense ablation minus one)</i>						
– Dim	✓	✓	—	✓	✓	—
– Sym	✓	✓	✓	—	✓	—
– Cons	✓	✓	✓	✓	—	—

Table 13: Sonnet 4.5 strict accuracy on Estonian Physics Olympiad problems by organizer-issued native difficulty (n=131). The curve is near-monotonically decreasing in difficulty and hits a hard floor of 0% at difficulties 3, 6, 8, and 10. Non-monotone bumps at 4 and 5 are within sampling noise on small per-bin counts.

Difficulty	n	Correct	Acc.
1	17	10	58.8%
2	15	3	20.0%
3	8	0	0.0%
4	12	3	25.0%
5	27	10	37.0%
6	9	0	0.0%
7	14	3	21.4%
8	9	0	0.0%
9	15	3	20.0%
10	5	0	0.0%
All	131	32	24.4%

Table 14: Cross-lingual Sonnet performance on the Estonian Physics Olympiad bilingual subset (n=59, identical problems, same judge protocol). Strict% = numeric/symbolic match; Liberal% = mean LLM-judge score (correct = 1, partial = 0.5, incorrect/unjudgeable = 0).

Language	n	Correct	Partial	Incorrect	Unjudgeable	Strict%	Liberal%
English (translated)	59	8	8	43	0	13.6%	20.3%
Estonian (original)	59	18	9	27	5	30.5%	38.1%

Table 15: Per-problem agreement matrix for the EN/ET cross-lingual ablation (n=59). The 4.3:1 asymmetry in the off-diagonal cells (ET-correct/EN-wrong vs. EN-correct/ET-wrong) rules out a noise explanation.

	ET correct	ET wrong
EN correct	5 (both correct)	3 (EN only)
EN wrong	13 (ET only)	38 (both wrong)